

Article

A Combined Deep Learning GRU-Autoencoder for the Early Detection of Respiratory Disease in Pigs Using Multiple Environmental Sensors

Jake Cowton ^{1,2,*}, Ilias Kyriazakis ² , Thomas Plötz ^{3,†}  and Jaume Bacardit ¹ ¹ Interdisciplinary Computing and Complex BioSystems (ICOS) Research Group, School of Computing, Newcastle University, Newcastle upon Tyne NE1 7RU, UK; jaume.bacardit@newcastle.ac.uk² Agriculture, School of Natural and Environmental Sciences, Newcastle University, Newcastle upon Tyne NE1 7RU, UK; ilias.kyriazakis@newcastle.ac.uk³ Open Lab, Newcastle University, Newcastle upon Tyne NE1 7RU, UK; thomas.ploetz@gatech.edu

* Correspondence: j.cowton2@newcastle.ac.uk

† Current address: School of Interactive Computing, Georgia Institute of Technology, Atlanta, GA 30318, USA.

Received: 22 June 2018; Accepted: 31 July 2018; Published: 2 August 2018



Abstract: We designed and evaluated an assumption-free, deep learning-based methodology for animal health monitoring, specifically for the early detection of respiratory disease in growing pigs based on environmental sensor data. Two recurrent neural networks (RNNs), each comprising gated recurrent units (GRUs), were used to create an autoencoder (GRU-AE) into which environmental data, collected from a variety of sensors, was processed to detect anomalies. An autoencoder is a type of network trained to reconstruct the patterns it is fed as input. By training the GRU-AE using environmental data that did not lead to an occurrence of respiratory disease, data that did not fit the pattern of “healthy environmental data” had a greater reconstruction error. All reconstruction errors were labelled as either normal or anomalous using threshold-based anomaly detection optimised with particle swarm optimisation (PSO), from which alerts are raised. The results from the GRU-AE method outperformed state-of-the-art techniques, raising alerts when such predictions deviated from the actual observations. The results show that a change in the environment can result in occurrences of pigs showing symptoms of respiratory disease within 1–7 days, meaning that there is a period of time during which their keepers can act to mitigate the negative effect of respiratory diseases, such as porcine reproductive and respiratory syndrome (PRRS), a common and destructive disease endemic in pigs.

Keywords: anomaly detection; deep learning; sensors; GRU; pig; health; disease

1. Introduction

The forecasting of an oncoming health challenge in animals is fundamental to maintaining a high level of health and animal welfare. The earlier a disease is predicted, the sooner it can be dealt with, thus lowering the overall impact of the disease on both the animal and the farm, and increasing the likelihood of treatment success [1]. Respiratory diseases, such as porcine reproductive and respiratory syndrome (PRRS), pneumonia, and pleurisy, are some of the most common types of disease found in commercial pig populations [2]. Therefore, it is highly valuable to be able to predict occurrences of respiratory disease, which we define as the point in time where respiratory disease prevalence increases from 0 to greater than 0.

There are many factors that influence the contraction of respiratory diseases, in particular, the environment which the pigs inhabit. A number of the respiratory disease risk factors discussed by Stark et al. [2]—specifically, air quality, temperature, and humidity—are now able to be monitored

文章一种深度学习GRU自动编码器组合用于猪的早期呼吸系统疾病检测，采用多种环境传感器

Jake Cowton ^{1,2,*}, Ilias Kyriazakis ² , Thomas Plötz ^{3,†}  and Jaume Bacardit ¹ ¹ 跨学科计算与复杂生物系统研究组，计算机学院，纽卡斯尔大学，纽卡斯尔 NE17RU，英国；jaume.bacardit@newcastle.ac.uk² 农业学院，自然与环境科学学院，纽卡斯尔大学，纽卡斯尔 NE1 7RU，英国； ilias.kyriazakis@newcastle.ac.uk³ 开放实验室，纽卡斯尔大学，纽卡斯尔 NE17RU，英

国； thomas.ploetz@gatech.edu* 通讯作者： j.cowton2@newcastle.ac.uk

† 当前地址：交互计算学院，佐治亚理工学院，亚特兰大，GA 30318，美国

收到：2018年6月22日；接受：2018年7月31日；发表：2018年8月2日



摘要：我们设计并评估了一种无假设的、基于深度学习的动物健康监测方法，特别针对基于环境传感器数据的生长猪呼吸系统疾病的早期检测。该方法使用了两个包含门控循环单元（GRU）的循环神经网络（RNN），用于构建一个自动编码器（GRU-AE），将来自各种传感器的环境数据输入其中以检测异常。自动编码器是一种经过训练以重建其输入模式的网络。通过使用不会导致呼吸系统疾病发生的环境数据训练GRU-AE，不符合“健康环境数据”模式的数据会产生更大的重建误差。所有重建误差均使用基于阈值的异常检测方法（该检测方法经过粒子群优化算法PSO优化）进行标注，从而触发警报。GRU-AE方法的结果优于最先进技术，当预测结果与实际观察结果出现偏差时，会触发警报。结果表明，环境变化可在1-7天内导致猪出现呼吸系统疾病症状，这意味着在猪出现呼吸系统疾病（如猪繁殖与呼吸综合征（PRRS），这是一种在猪中常见且具有破坏性的地方性疾病）的负面影响之前，其饲养者有一段时间可以采取行动减轻这种影响。

关键词：异常检测；深度学习；传感器；门控循环单元；猪；健康；疾病

1. 引言

在动物中预测即将到来的健康挑战是维持高水平健康和动物福利的基础。越早预测疾病，就能越早处理，从而降低疾病对动物和农场整体的影响，并提高治疗成功的可能性 [1]。猪繁殖与呼吸综合征（PRRS）、肺炎和胸膜炎等呼吸系统疾病是商业猪群中最常见的疾病类型之一 [2]。因此，能够预测呼吸系统疾病的发生具有极高的价值，我们将其定义为呼吸系统疾病患病率从0增加到大于0的时间点。

影响呼吸系统疾病发生发展的因素有很多，特别是猪所生活的环境。Stark 等人讨论的呼吸系统疾病风险因素中， [2]——具体包括空气质量、温度和湿度——现在能够通过廉价的电子传感器进行实时、高分辨率连续监测

continuously in real time and at a high resolution using inexpensive electronic sensors. Such data can be used to further understand the relationship between environmental conditions and respiratory disease.

In most situations, recording data continuously means storing data that predominantly describes normal circumstances, as anomalous events are less frequent. The large amounts of normal data generated allow us to train a solid model of normality. Hence, it is better to model normality and then quantify any deviations from it.

The objective of this paper is to describe and evaluate a methodology for the early warning of anomalies pertaining to respiratory disease in growing pigs, based on environmental factors. A recurrent autoencoder, built using deep learning (gated recurrent units (GRU)), learns how to reconstruct raw sensor data which does not lead to occurrences of respiratory disease within a batch of pigs. This GRU-autoencoder (GRU-AE) can then be used to “score” a window of sensor data according to how close it is to normality. A threshold-based anomaly detector that was optimised using particle swarm optimisation (PSO) used this score to determine whether to raise an alert for scores which are too great and therefore represent environmental conditions that do not fit in line with the data it was trained to represent.

The benefits of using such a method are that there is no need for hand-crafted features, as is typically required in classical statistical approaches, meaning that raw sensor data can be used with minimal preprocessing. Also, no arbitrary thresholds are set, as the detection of anomalous events is wholly data-driven. In addition, there is no requirement for an in-depth understanding of the relationship between environmental conditions and pig health, as this is handled implicitly within the network. This work is structured as an anomaly detection solution, rather than, for example, predicting increases in respiratory disease prevalence; therefore, the method is able to perform on data consisting of low disease prevalence without affecting performance. However, the more cases of disease there are, and the higher the prevalence at these times, the easier it is to fine-tune and validate the method.

Malhotra et al. [3] demonstrated the potential of a long short-term memory (LSTM)-based encoder-decoder structure for multi-sensor anomaly detection, showing the methodology has good potential for using data which is not easily predictable [4]. This implementation used LSTM cells, rather than a standard artificial neural network (ANN) structure, in order to handle the temporal dimension, and used reconstruction error was used to determine anomalies. However, the results of their implementation on various datasets suffered distinctly either in precision or recall (defined in Section 4.1.1). This paper revisits the design choices made in previous work, such as that by Malhotra et al., specifically by:

- exchanging long short-term memory cells for GRU cells
- introducing particle swarm optimisation to optimise an anomaly detector that determines whether the loss from the GRU-AE is anomalous
- changing how the overall system is evaluated to demonstrate a more balanced anomaly detection system

The remainder of the paper is structured as follows. Firstly, Section 2 gives context for the approach taken. Section 3 explains the composition of the data and the preprocessing that was applied, followed by Section 4, which details how each of the models were constructed. Section 5 outlines the performance of the systems described, followed by the discussion in Section 6 and the conclusions drawn in Section 7.

2. Background

In recent years, agricultural research has begun to shift away from relying only on classical statistical approaches and started to regularly incorporate machine learning (ML) methodologies, such as Support Vector Machines (SVM) and Random Forest Classifiers, into the approach to data analysis [5–10].

Recent work has seen SVMs being used, for example, to detect anomalies in the production curve of eggs produced by commercial hens, covering a range of prediction windows for the purpose

这些数据可用于进一步理解环境条件与呼吸系统疾病之间的关系。

在大多数情况下，连续记录数据意味着存储大量主要描述正常情况的数据，因为异常事件频率较低。生成的正常数据量很大，使我们能够训练一个坚实的正常模型。因此，最好先对正常性进行建模，然后量化任何偏离正常性的情况。

本文的目的是描述和评估一种基于环境因素，对生长猪呼吸系统疾病进行早期预警的方法。一个使用深度学习（门控循环单元（GRU））构建的循环自动编码器，学习如何重建一批猪中不会导致呼吸系统疾病的发生的原始传感器数据。然后，该GRU自动编码器（GRU-AE）可以根据其与正常性的接近程度对传感器数据窗口进行“评分”。一个使用粒子群优化（PSO）优化的基于阈值的异常检测器使用此分数来确定是否对分数过高的情况发出警报，因为这些分数代表的环境条件不符合其训练所代表的数据。

使用这种方法的优点在于，无需手工设计特征，正如经典统计方法通常所要求的，这意味着原始传感器数据可以在最少的预处理下使用。此外，不会设置任意阈值，因为异常事件的检测完全是数据驱动的。另外，不需要深入理解环境条件与猪健康之间的关系，因为这一点在网络内部被隐式处理。这项工作被构建为一个异常检测解决方案，而不是例如预测呼吸系统疾病患病率的增加；因此，该方法能够在由低患病率数据组成的集合上运行，而不会影响性能。然而，病例越多，以及这些时候的患病率越高，就越容易微调和验证该方法。

Malhotra等人 [3] 展示了基于长短期记忆单元（LSTM）的编码器-解码器结构在多传感器异常检测中的潜力，表明该方法对于使用难以预测的数据具有良好的潜力 [4]。该实现使用了长短期记忆单元，而不是标准的人工神经网络（ANN）结构，以便处理时间维度，并使用重建误差来确定异常。然而，他们在各种数据集上的实现结果在精确率或召回率（定义在Section 4.1.1）上明显受损。本文重新审视了先前工作中的设计选择，特别是Malhotra等人所做的工作，具体包括：

- 用门控循环单元（GRU）细胞替换长短期记忆（LSTM）细胞
- 将粒子群优化算法引入以优化异常检测器，该检测器用于判断GRU-AE的损失是否异常

- 改变整体系统的评估方式，以展示更均衡的异常检测系统

本文的其余部分结构如下。首先，第2 节介绍了所采用方法的背景。第3 节解释了数据的组成以及所应用的预处理，随后是第4节，该节详细介绍了每个模型的构建方法。第5 节概述了所描述系统的性能，随后是第6 节的讨论以及第7节得出的结论。

2. 背景

近年来，农业研究开始逐渐摆脱单纯依赖经典统计方法，转而定期将机器学习（ML）方法，如支持向量机（SVM）和随机森林分类器，纳入数据分析方法中 [5–10]。

近期研究已看到支持向量机被用于检测商业母鸡产蛋量生产曲线中的异常，覆盖了多种预测窗口，以实现早期预警系统的目的

of an early warning system [5]. Other research has applied ensemble learning methods (combined models) for the prediction of avian influenza prevalence using meteorological data, showing improved performance over support vector regression [6].

Classification methods have been used to separate normality from abnormality through the implementation of one-class classifiers (OCCs). SVM-based methods are very common in this approach, though there are many potential methods which can be applied. Models of this type usually have high sensitivity but a comparatively poor specificity [8], which is to be expected given the natural imbalance in the datasets that these methodologies are commonly used for. However, these metrics are not always suitable, particularly if the true negatives drastically outweigh the number of false positives. Cabral et al. compared a number of common OCCs with the proposal of features boundary detection for one-class classification (FBDOCC) of heart disease [8].

There are also approaches to anomaly detection outside of classification, such as clustering [11] and correlation analysis [12]. Clustering uses distance metrics to determine if a new point's distance from a cluster centroid exceeds an acceptable value: if this is the case, it is considered anomalous. However, methods falling into this category are typically unable to strike a reasonable balance between quality of analysis and speed due to the innate characteristics of time-series data [13]. Correlation analysis uses similarity metrics to determine if new data is similar to known normal data.

Anomaly detection can be used in health applications to detect outbreaks of disease or other deviations from normality. A common methodology used for this task are Bayesian networks, which have been shown to outperform classical statistical methods, such as control charts, moving average models, and ANOVA regression [14]. In one simulation of an anthrax outbreak [15], the Bayesian network was able to correctly trigger an alarm within 2 days, in comparison to 9 days for the moving average model and 11–12 days for the others. Anomaly detection is very well suited to healthcare, as often there are not necessarily certain conditions that constitute a problem, but rather deviations from certain conditions. For example, when detecting arrhythmia through electrocardiography (ECG), it is not possible to have knowledge of all of the different types, as new types can occur in the future [16]. It is therefore necessary to understand what normality looks like, rather than what problems look like.

In health applications where sensors are used, there are two main challenges. Firstly, there is a need to focus not only on a high detection rate, but also a low false positive rate [17], as, if this is high, alerts from this system will not be trusted. Secondly, though not in all cases, anomalies ought to be detected in near real time [18] in order to provide a quick response so that interventions can be carried out. This creates a substantial limitation on which methods can be used, as some are very resource intensive.

Often, these approaches require a transformation be carried out on the data in order to generate a useful feature representation [19–21]. However, deep learning-based methods benefit by not requiring this step. Raw data can be passed to a deep network without any hand-crafting of features, as this is handled intrinsically by the network [16]. However, this requires a substantial amount of data to function effectively [22]. This is particularly valuable for high-dimensional data, as, provided the network is deep enough to model the data, all data can be utilised.

A relatively small amount of work in agriculture has made use of emerging technologies, such as deep learning, which mostly focuses on image analysis and remote sensing [23]. This paper demonstrates how deep learning methods, used in conjunction with evolutionary algorithms, can be used for time-series analysis for the purpose of anomaly detection.

的早期预警系统 [5]。其他研究则应用集成学习方法（组合模型）利用气象数据预测禽流感流行情况，显示出比支持向量回归更好的性能 [6]。

分类方法已用于通过单类分类器（OCCs）实现正常与异常的分离。基于支持向量机的方法在此方法中非常常见，尽管有许多潜在方法可以应用。此类模型的灵敏度通常很高，但特异性相对较差 [8]，考虑到这些方法通常用于的数据集的自然不平衡，这是可以预料的。然而，这些指标并不总是适用，特别是当真实负例的数量远远超过假正例的数量时。Cabral等人比较了多种常见的OCCs，并提出了用于单类分类（FBDOCC）心脏病 [8]的特征边界检测方法。

还有分类之外的异常检测方法，如聚类 [11]和相关分析 [12]。聚类使用距离度量来确定新点与聚类质心的距离是否超过可接受值：如果是这样，则被视为异常。然而，属于此类的方法通常无法在分析质量和速度之间取得合理平衡，这是由于时间序列数据的固有特性 [13]。相关分析使用相似性度量来确定新数据是否与已知正常数据相似。

异常检测可用于健康应用，以检测疾病爆发或其他偏离正常的情况。用于此任务的一种常见方法是贝叶斯网络，研究表明其性能优于经典统计方法，例如控制图、移动平均模型和ANOVA回归 [14]。在一项炭疽疫情模拟中 [15]，贝叶斯网络能够在2天内正确触发警报，而移动平均模型需要9天，其他方法则需要11-12天。异常检测非常适合医疗保健领域，因为通常不存在构成问题的特定条件，而是偏离某些条件。例如，通过心电图（ECG）检测心律失常时，不可能了解所有不同类型，因为未来可能会出现新类型 [16]。因此，有必要了解正常状态是什么样子，而不是问题是什么样子。

在传感器用于健康应用的情况下，存在两个主要挑战。首先，不仅需要关注高检测率，还需要关注低误报率 [17]，因为如果误报率高，系统发出的警报将不被信任。其次，虽然并非所有情况下都如此，但异常检测应在近乎实时 [18] 进行，以便快速响应，从而采取干预措施。这给可使用的带来了重大限制，因为某些方法非常资源密集。

通常，这些方法需要对数据进行转换，以便生成有用的特征表示 [19–21]。然而，基于深度学习的方法则无需这一步骤。原始数据可以直接输入深度网络，无需手工设计特征，因为特征提取是网络内在功能的一部分 [16]。但这也需要大量数据才能有效运行 [22]。对于高维数据而言，这一点尤其有价值，只要网络足够深能够对数据进行建模，所有数据都能被利用。

农业领域相对较少的工作利用了新兴技术，如深度学习，这些技术主要关注图像分析和遥感 [23]。本文展示了深度学习方法与进化算法结合如何用于时间序列分析，目的是进行异常检测。

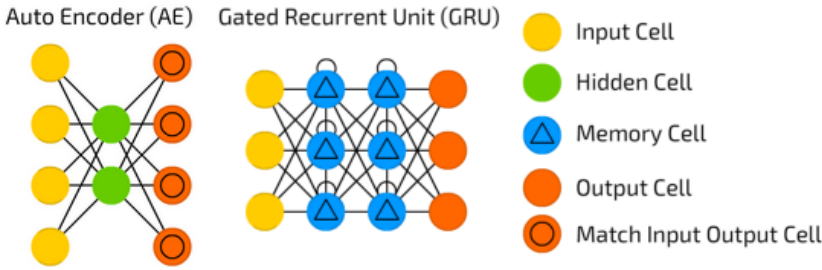


Figure 1. A visualisation of a classic autoencoder (left) and recurrent neural network consisting of gated recurrent units (GRUs) (right) [24]. These two methodologies are what we combined to create an autoencoder (AE) capable of utilising temporality.

Autoencoders are artificial neural networks (ANNs) that are trained to reconstruct input patterns (Figure 1). Given an input tensor N , an encoding M is found which is used to create N' . The network is trained to minimise the loss between N and N' . The transformation of N to M is the encoder, and the attempted reconstruction of M to N' constitutes the decoder. Autoencoders have shown good performance in assisting anomaly detection, often through learning the feature representation M and using it as the input to an OCC [25–27], though some papers have made use of the reconstruction error to detect anomalous data [28].

3. Experimental Design

3.1. Data Collection

Data was collected for growing pigs from a variety of farms across Europe as part of an EU-funded multidisciplinary project. Each farm was represented by a number of batches of pigs (3–16 batches), where a batch consists of varying population sizes (40–1294 pigs), and the batches were monitored for varying lengths of time (14–145 days). The starting weight of the pigs was variable, ranging from 17.5 to 27.6 kg. The number of pigs per batch affected by respiratory disease was recorded on a daily basis. Respiratory disease prevalence was divided into four levels of severity, as defined by Zoetis’s Individual Pig Care [29]. A pig could be either scored (identified as diseased but no treatment given) or treated for each level of severity. Health data regarding the number of pigs showing symptoms of disease was collected by a trained person. They manually counted the number of pigs in each batch that showed symptoms of certain diseases and classified it by the level of severity into which the symptoms fell. Given the nature of the system, pigs cannot be identified individually each day, so the number of pigs that are classified as showing symptoms can be new cases or pigs that were identified as such in the previous day.

The environmental data was collected using General Alert [30] sensors, which were suspended in the centre of the room in which the batch was located. The sensors recorded the environmental features relevant to pig health [31]. Briefly, the system monitored temperature (a sensor range of $-50\text{ }^{\circ}\text{C}$ to $+250\text{ }^{\circ}\text{C} \pm 0.05\%$), relative humidity (a sensor sensitivity of $\pm 2\%$ and operating temperature range of $-40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$), and CO_2 concentration (with a range from 0 ppm to 5000 ppm ± 30 ppm or $\pm 3\%$ of reading, operating temperature range from $0\text{ }^{\circ}\text{C}$ to $50\text{ }^{\circ}\text{C}$, and humidity from 0% to 95%), which was sampled every minute.

3.2. Data Preprocessing

Outliers were determined to be anything outside $\mu \pm 2\sigma$ per variable per batch and were removed from the data. Missing data induced both from the removal of outliers and due to hardware issues were filled using linear interpolation. This was deemed acceptable, as the environment has low short-term variance, meaning that, on average, the assumption of a linear change between two values was reasonable. The results of this preprocessing of sensor data are shown in Figure 2. The three sensor

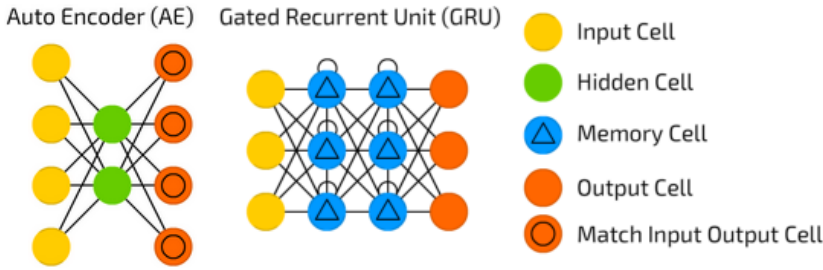


图1. 一个经典自编码器（左）和由门控循环单元（GRU）组成的循环神经网络（右）的可视化 [24]。这两种方法是我们将它们结合在一起以创建一个能够利用时序性的自编码器（AE）。

自编码器是人工神经网络（ANNs）的一种，它被训练用于重建输入模式（图 1）。给定一个输入张量 N ，找到一个编码 M ，用于创建 N' 。该网络被训练以最小化 N 和 N' 之间的损失。将 N 转换为 M 是编码器，而将 M 尝试重建为 N' 则构成解码器。自编码器在辅助异常检测方面表现出良好性能，通常通过学习特征表示 M 并将其用作OCC [25–27]，的输入——尽管一些论文利用重建误差来检测异常数据 [28]。

3. 实验设计

3.1. 数据收集

数据是作为欧盟资助的多学科项目的一部分，从欧洲各地的多种农场收集的用于生长猪。每个农场都由一批或多批猪（3-16批）代表，一批猪的种群规模各不相同（40-1294头），并且这些批次被监测了不同的时间长度（14-145天）。猪的初始体重是可变的，范围从17.5到27.6公斤。每批猪中受呼吸系统疾病影响的猪的数量每天都会被记录。呼吸系统疾病的发病率被分为四个严重程度级别，如Zoetis的个体猪护理 [29]所定义。一头猪可以被评分（被确定为患病但未给予治疗）或被治疗，具体取决于其严重程度级别。与健康相关的数据，即表现出疾病症状的猪的数量，是由一名训练有素的人员收集的。他们手动统计了每批中表现出某些疾病症状的猪的数量，并根据症状所属的严重程度级别进行分类。鉴于系统的性质，每天无法识别每头猪的个体身份，因此被归类为表现出症状的猪的数量可能是新病例，也可能是前一天被确定为这样的猪。

环境数据是使用通用警报 [30] 传感器收集的，这些传感器悬挂在存放批次的房间中央。传感器记录了与猪健康相关的环境特征 [31]。简而言之，该系统监测温度（传感器范围为 $-50\text{ }^{\circ}\text{C}$ 至 $+250\text{ }^{\circ}\text{C} \pm 0.05\%$ ）、相对湿度（传感器灵敏度为 $\pm 2\%$ ，工作温度范围为 $-40\text{ }^{\circ}\text{C}$ 至 $+85\text{ }^{\circ}\text{C}$ ），以及 CO_2 浓度（范围从 0 ppm 至 5000 ppm ± 30 ppm 或 $\pm 3\%$ 的读数，工作温度范围为 $0\text{ }^{\circ}\text{C}$ 至 $50\text{ }^{\circ}\text{C}$ ，湿度从 0% 至 95%），这些数据每分钟采样一次。

3.2. 数据预处理

异常值被定义为任何超出 $\mu \pm 2\sigma$ 每个变量每个批次范围之外的数据，并从数据中移除。由于移除异常值和硬件问题导致的缺失数据均使用线性插值填充。这被认为是可接受的，因为环境具有低短期方差，这意味着平均而言，两个值之间的非线性变化假设是合理的。传感器数据的预处理结果如图2所示。三个传感器

readings—temperature, humidity, and CO₂—constitute the inputs to the models outlined in Section 4. The four severity levels were summed and used to calculate the proportion of pigs showing symptoms of respiratory disease on a given day, rather than looking at one specific level of severity (Equation (1)).

$$\text{prevalence} = \frac{\text{Number of pigs showing any symptoms}}{\text{Total number of pigs}}$$

(1)

This was done because respiratory diseases rarely exceed the classification of light or mild, as in most cases they are treated before the disease develops into more severe symptoms. Therefore, there were very few cases of severe or irrecoverable respiratory disease across all farms. Even with the aggregation of data, the level of prevalence was still very low, and it decreased even further over time (Figure 3). The sensor data was merged with the daily health data by upscaling the resolution of the latter by duplicating the values for the full 24 h period. All input was normalised to between 0 and 1.

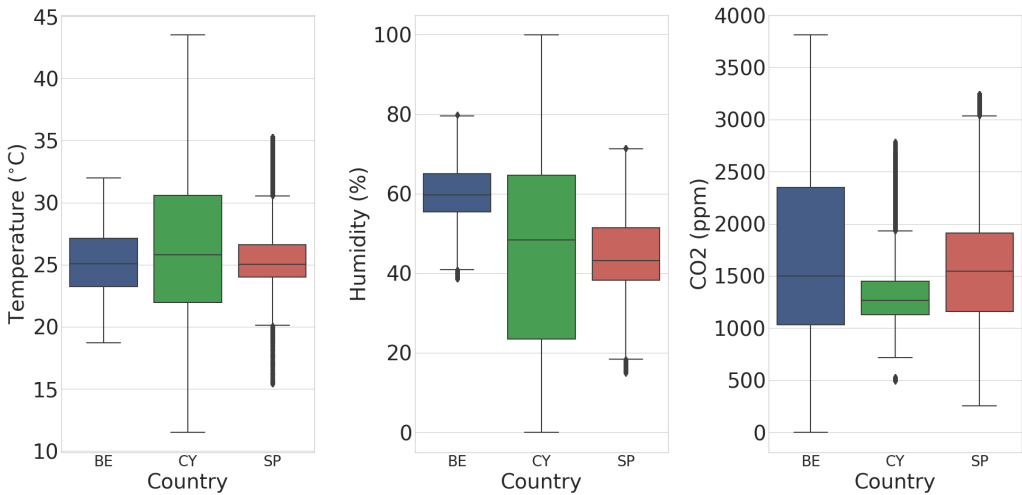


Figure 2. A boxplot describing the temperature, humidity, and CO₂ sensor data for the three countries included in the data collection—Belgium (BE), Cyprus (CY), and Spain (SP)—after preprocessing.

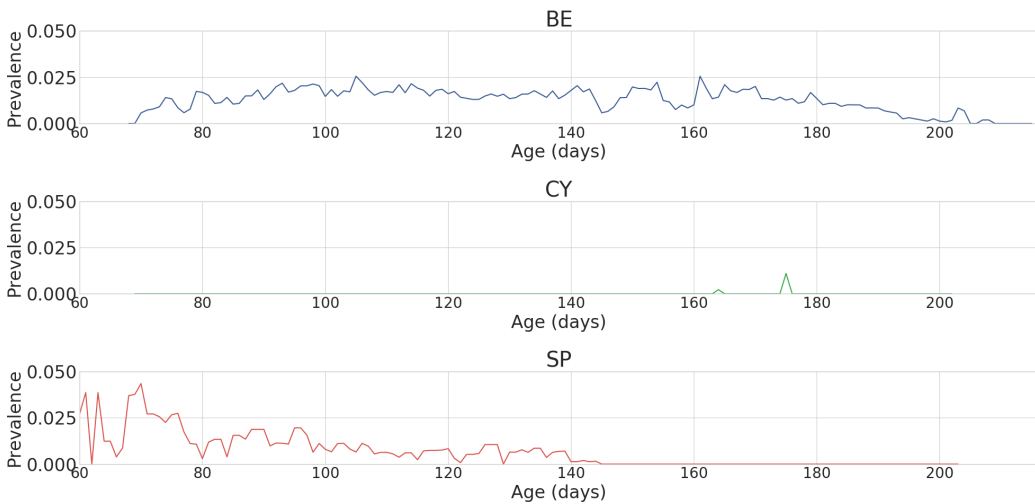


Figure 3. A graph showing the average prevalence of respiratory disease against age (in days) in all batches of growing pigs in Belgium (BE), Cyprus (CY), and Spain (SP) after preprocessing.

读数——温度、湿度和CO₂——构成第4节中概述的模型的输入。四种严重程度被汇总并用于计算在特定一天内表现出呼吸系统疾病症状的猪的比例，而不是查看特定的严重程度级别（公式（1））。

$$\text{prevalence} = \frac{\text{Number of pigs showing any symptoms}}{\text{Total number of pigs}}$$

(1)

这是因为在大多数情况下，呼吸系统疾病在发展成更严重的症状之前就会被治疗，因此呼吸系统疾病很少超过轻症或中症的分类。所以，在所有农场中，严重或无法恢复的呼吸系统疾病病例非常少。即使合并了数据，患病率仍然非常低，并且随着时间的推移进一步下降（图3）。传感器数据通过将后者（每日健康数据）的分辨率上采样，通过复制值来合并，以覆盖完整的24小时周期。所有输入数据都归一化到0到1之间。

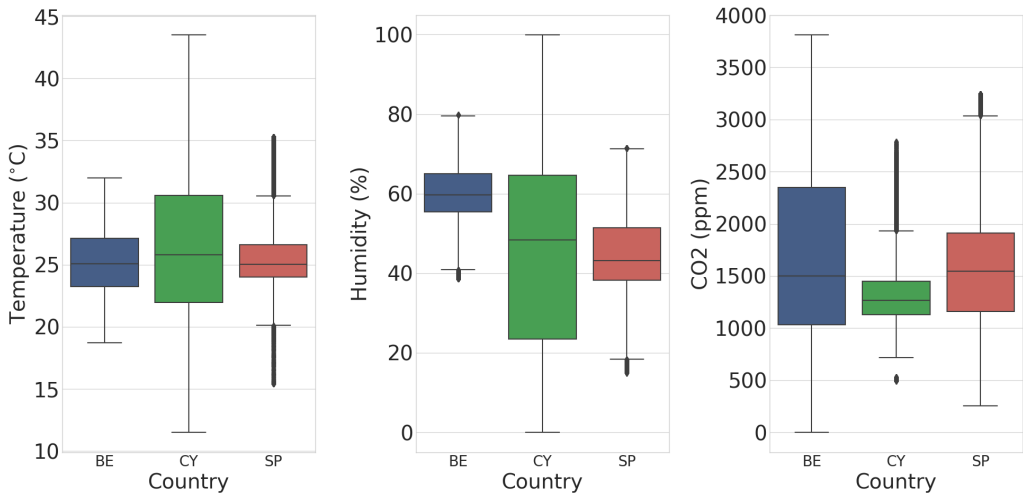


图2。 一个箱线图，描述了温度、湿度和CO₂ 传感器数据，针对数据收集中包含的三个国家——比利时（BE）、塞浦路斯（CY）和西班牙（SP）——在预处理后的情况。

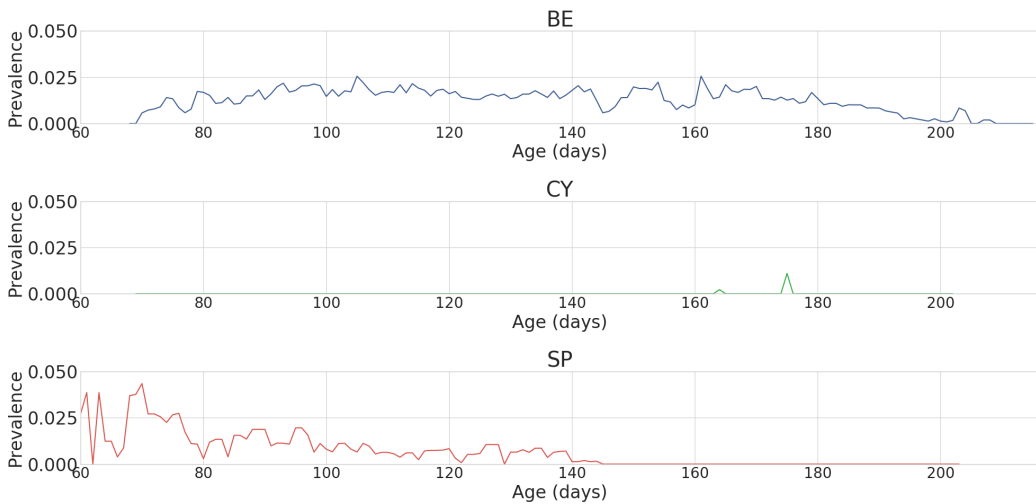


图3。 一个图表，显示了呼吸系统疾病在比利时（BE）、塞浦路斯（CY）和西班牙（SP）所有生长猪批次的平均患病率与年龄（以天为单位）的关系，在预处理后。

3.2.1. Separation of Assumed Healthy and Assumed Unhealthy Data

As this paper investigates the effects of environment on health, it was necessary to account for the existence of a lag period between a change in environment and a change in health, as a change in environment will not immediately result in an outbreak of disease [32]. Lag periods of 14 and 21 days were both evaluated [32]; greater windows were not used, as this would greatly reduce the amount of data available for training. This lag period was referred to as the assumed unhealthy window, u . Using this window, the data was labelled as one of two sets: “assumed healthy” and “assumed unhealthy”. Data from any point where the proportion of pigs showing symptoms was greater than 0 to u days after the disease prevalence decreases to 0 was labelled “assumed unhealthy”; all other data was labelled “assumed healthy”. All data at this point was broken down into 30 min “frames” using a sliding window.

3.2.2. Splitting Data into Train, Validation, and Test Sets

It was necessary to split data into training and testing sets. Leave-one-batch-out cross-validation was used to achieve this (Figure 4). Our method, detailed in Section 4.1, and the GRU-regression (GRU-R) model used for comparison, detailed in Section 4.3.3, were each composed of two parts: a GRU network and a threshold-based anomaly detector. Both the GRU network and anomaly detector required two sets of data for the full workflow. H_t, H_v were used for the GRU network, and O_e and O_t were used for the anomaly detector. H_t was used for the training of GRU networks and H_v was used to validate it. O_e was used to optimise an anomaly detector via PSO and O_t was the set the anomaly detector was tested on to attain an evaluation metric. There was no overlap between any of these sets.

O_t was a single, independent farm batch, O_e was composed of three randomly chosen batches, H_v contained only the assumed healthy data of another independent farm batch, and H_t contained only the assumed healthy data from all the remaining farm batches.

Each of the four datasets could be represented as a matrix $X \in \mathbb{R}^{n \times 3}$, where n is the number of minutes of data in a farm batch, and each X_n contains sensor data for temperature, humidity, and CO_2 . This matrix was broken down into frames using a sliding window of size w , which moved one timepoint (1 min) per step, creating $T \in \mathbb{R}^{m \times w \times 3}$, where m was the number of frames in the set.

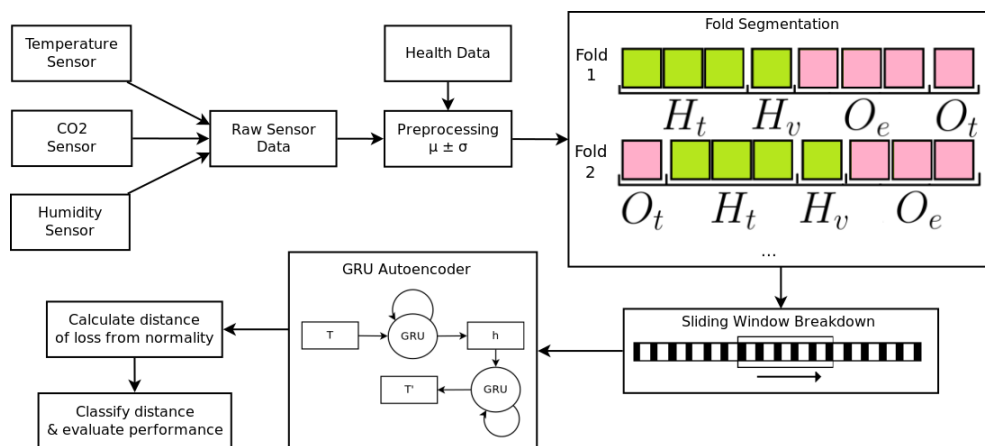


Figure 4. An overview of the entire processing workflow, from the raw sensor data to anomaly detection. Temperature, CO_2 , and humidity sensors were preprocessed and merged with daily health data. This was then broken down into four folds— H_t and H_v (used for the GRU-AE), and O_e and O_t (used for the anomaly detection). Each of these folds were broken down using a sliding window sized at 30 min. Each window of 30 min was processed by the GRU-AE, and a reconstruction error was produced. The distance between this reconstruction error and normality was used to detect whether the window is anomalous or not, using a threshold-based anomaly detector optimised using particle swarm optimisation.

3.2.1. 假设健康数据与假设异常数据的分离

由于本文研究的是环境对健康的影响，因此有必要考虑环境变化与健康变化之间存在滞后期，因为环境变化不会立即导致疾病爆发 [32]。评估了14天和21天的滞后期 [32]；未使用更长的窗口，因为这会大大减少可用于训练的数据量。这个滞后期被称为假设不健康窗口 u 。使用这个窗口，数据被标记为两个集合之一：“假设健康”和“假设不健康”。当症状出现比例大于0到 u 天后的疾病患病率下降到0时，该点的数据被标记为“假设不健康”；所有其他数据被标记为“假设健康”。此时所有数据都使用滑动窗口分解为30分钟的“帧”。

3.2.2. 将数据分割为训练集、验证集和测试集

需要将数据分割为训练集和测试集。采用留一批次交叉验证方法实现（图4）。我们的方法，详见第4.1节，以及用于对比的GRU回归（GRU-R）模型，详见第4.3.3节，均由两部分组成：GRU网络和基于阈值的异常检测器。GRU网络和异常检测器在完整工作流程中均需要两组数据。 H_t, H_v 用于GRU网络，而 O_e 和 O_t 用于异常检测器。 H_t 用于GRU网络的训练，而 H_v 用于验证它。 O_e 用于通过粒子群优化算法优化异常检测器，而 O_t 是用于获得评估指标的异常检测器测试集。这些集合之间没有任何重叠。

O_t 是一个单一、独立的农场批次， O_e 由三个随机选择的批次组成， H_v 仅包含另一个独立农场批次的假定健康数据，而 H_t 仅包含所有剩余农场批次的假定健康数据。

每个数据集都可以表示为一个矩阵 $X \in \mathbb{R}^{n \times 3}$ ，其中 n 是农场批次中数据分钟数，而每个 X_n 包含温度、湿度和 CO_2 的传感器数据。该矩阵使用大小为 w 的滑动窗口分解为帧，每次移动一个时间点（1分钟），创建 $T \in \mathbb{R}^{m \times w \times 3}$ ，其中 m 是集合中的帧数。

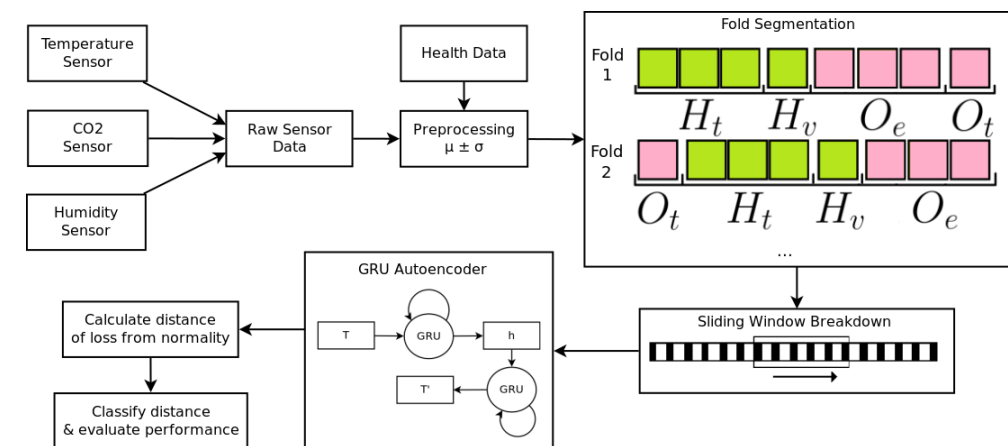


图 4. 整个处理工作流程的概述，从原始传感器数据到异常检测。温度、 CO_2 和湿度传感器被预处理并合并到每日健康数据中。然后将其分解为四个折叠—— H_t 和 H_v (用于GRU-AE)，以及 O_e 和 O_t (用于异常检测)。这些折叠中的每一个都使用30分钟的滑动窗口进行分解。每个30分钟的窗口由GRU-AE处理，并产生一个重建误差。这个重建误差与正常值的距离被用于检测窗口是否异常，使用一个基于阈值的异常检测器，该检测器使用粒子群优化进行优化。

3.2.3. Preliminary Experiments Utilising Batch Normalisation

Batch normalisation is a deep learning technique that has shown to be very effective at training robust networks by performing a per layer normalization process on input data. Please note that the meaning of batch in this context differs from the farm batches used throughout the rest of the manuscript. In this context, batches are the subsets of the training data that are fed in blocks to the network being trained with the subsequent back-propagation-based error correction. In anomaly detection, the effect of this technique is not necessarily desirable and, in this particular application, based on preliminary experiments, was found to be detrimental to performance. This was because batch normalisation allows a model to deal with internal covariate shift by normalising each mini-batch by its mean and variance [33], meaning that the model is more capable of handling data it has never seen before. The decrease in performance when using batch normalisation showed that there is a requirement that the model be highly sensitive to slight variations in the data, and so it was not used in the final models.

4. Time-Series Early Warning Methods

4.1. GRU-Autoencoder

The implementation of an autoencoder model (Figure 4) consisted of two multi-layer GRU networks—an encoder and a decoder—that were trained simultaneously. This is instead of the traditional ANN structure (Figure 1), which does not intrinsically account for temporality in data. The encoder learned a fixed-size representation h of an input T . The quality of this representation was measured by using it to attempt a reconstruction of T using the decoder. The model was optimised to minimise the reconstruction error, calculated using binary cross-entropy loss, between T and T' . T was reversed when calculating loss, as it creates short-term dependencies that result in an increased performance in sequence-to-sequence networks [34].

H_t was used to train the GRU-AE using mini-batch RMSprop gradient descent [35]. This dataset contained only “assumed healthy” data, which resulted in the GRU-AE optimising towards a lower reconstruction error for this class of data and a higher reconstruction error for all other data. Hyper-parameters, hidden size, number of layers, mini-batch size, dropout, learning rate, and momentum were selected using grid search, and the parameters which resulted in the lowest reconstruction error for each fold’s H_v were selected. The hidden size represents the amount of information the GRU-AE can store for each time series it is given. As there was no requirement for a large amount of memory for this task, the hidden size representation identified by the grid search was expected to be fairly small. The number of recurrent layers represents the capability of the GRU to create deep models and understanding of the time series. These are expensive in terms of computation and memory. The mini-batch size is how many sequences are passed forward through the network before back-propagating the loss. This is typically desired to be as large as possible to speed up training, though is entirely dependent on the other parameters and the amount of memory available. Dropout is a regularisation technique used to reduce the likelihood of over-fitting and is common in standard artificial neural networks. The learning rate and momentum are also used in standard artificial neural networks. They determine how drastically weights are changed during back-propagation. The selected hyper-parameters were then used to calculate reconstruction errors for all frames in O_e and O_t .

Once all reconstruction errors were calculated, the threshold-based anomaly detector was used to determine whether each reconstruction error of a frame was to be considered anomalous or not. This was estimated by calculating the Mahalanobis distance (Equation (2)) between the reconstruction error of a given frame e and the distribution of the reconstruction errors for all frames of H_v , defined as R , where $\Sigma = COV(e, R)$. This was then smoothed using locally weighted smoothing (LOWESS) [36]. If the smoothed distance was above some threshold τ (the optimisation of which is described in Section 4.1.1), then it was considered an anomalous frame.

3.2.3. 批量归一化初步实验

批量归一化是一种深度学习技术，通过在输入数据上执行逐层归一化过程，已被证明在训练鲁棒网络方面非常有效。请注意，这里的批量与本文其余部分使用的农场批次含义不同。在这种情况下，批次是训练数据的小块，这些小块被分批输入到正在训练的网络中，随后通过基于反向传播的错误校正。在异常检测中，这种技术的影响不一定是理想的，并且在本次特定应用中，根据初步实验发现会损害性能。这是因为批量归一化允许模型通过对其均值和方差进行归一化来处理内部协变量偏移，这意味着模型更有能力处理它以前从未见过的新数据。使用批量归一化时性能下降表明，模型需要高度敏感于数据中的微小变化，因此它没有被用于最终模型。 [33], 这意味着模型更有能力处理它以前从未见过的新数据。使用批量归一化时性能下降表明，模型需要高度敏感于数据中的微小变化，因此它没有被用于最终模型。

4. 时间序列早期预警方法

4.1. GRU自动编码器

自动编码器模型（图 4）的实现由两个多层门控循环单元网络组成——一个编码器和一个解码器——它们被同时训练。这不同于传统的ANN结构（图 1），后者在数据中不内在地考虑时序性。编码器学习输入 h 的固定大小表示 T 。该表示的质量通过使用它来尝试使用解码器重建 T 来衡量。该模型被优化以最小化重建误差，该误差使用二元交叉熵损失计算， T 和 T' 。 T 在计算损失时被反转，因为它创建了短期依赖关系，这导致序列到序列网络性能提高 [34]。

H_t 被用于使用小批量RMSprop梯度下降训练GRU-AE [35]。该数据集仅包含“假设健康”数据，这导致GRU-AE针对此类数据优化出较低的重建误差，而针对所有其他数据则优化出较高的重建误差。超参数、隐藏层大小、层数、小批量大小、dropout、学习率和动量通过网格搜索选择，并选择了在每次折的<样式id='5'>H</样式>上重建误差最低的参数。隐藏层大小表示GRU-AE为每个给定的时间序列可以存储的信息量。由于这项任务对内存没有大量需求，网格搜索识别出的隐藏层大小预期会相对较小。层数表示GRU创建深度模型和理解时间序列的能力。这些在计算和内存方面都很昂贵。小批量大小是指多少个序列在通过网络进行反向传播之前被前向传递。这通常希望尽可能大以加快训练速度，尽管完全取决于其他参数和可用的内存量。Dropout是一种正则化技术，用于降低过拟合的可能性，在标准人工神经网络中很常见。学习率和动量也在标准人工神经网络中使用。它们决定了在反向传播期间权重变化的剧烈程度。然后使用选定的超参数计算 O_e 和 O_t 中所有帧的重建误差。

一旦所有重建误差都计算完毕，阈值异常检测器被用于判断每帧的重建误差是否应被视为异常。这通过计算给定帧的重建误差 e 与所有帧的重建误差分布 H_v ，即 R ，之间的马氏距离（公式（2））来估计 $COV(e, R)$ 。然后使用局部加权平滑（LOWESS） [36]对距离进行平滑处理。如果平滑后的距离高于某个阈值 τ (其优化方法在4.1.1节中描述)，则该帧被视为异常帧。

$$d = \sqrt{(e - \bar{R})^T \Sigma^{-1} (e - \bar{R})} \quad (2)$$

4.1.1. PSO-Optimised Anomaly Detection

Once all distances were calculated for all frames, the next step was to optimise the threshold-based anomaly detector, capable of determining which frames should be considered anomalous, using PSO [37]—a metaheuristic, which uses concepts of swarm intelligence to find a set of hyper-parameters that produce a near-optimal solution. Consider the vector of Mahalanobis distances of all frames in a sequence $\vec{D} = \{d_0, d_1, d_2, \dots, d_n\}$, such that each d_i is the Mahalanobis distance between the reconstruction error of a frame at index i from R . An alert was defined as any $d_i : d_{i-1} < \tau \wedge d_i \geq \tau$. The validity of an alert was assessed by determining whether disease prevalence increased from 0 within an acceptable forecasting window starting at $i + \alpha$ and ending at $\alpha + \beta$. If so, and if no other alert had been raised for this increase in prevalence, the alert was considered a true positive, else a false positive. Furthermore, consider the vector $\vec{F} = \{f_0, f_1, f_2, \dots, f_n\}$ such that each f_i contains the number of animals showing symptoms of respiratory disease at frame i . A false negative was determined to be any frame $i : f_{i-1} = 0 \wedge f_i > 0$ for which there is no alert within $i - \alpha$ and $i - \beta$.

The parameters τ , α , and β of the detector were the parameters to be optimised using PSO. The particles were optimised for each fold independently in O_e , and the best values for each parameter were then tested on O_t for a final performance evaluation. Each particle was randomly initialised with three dimensions representing τ , α , and β . The fitness function (Equation (3)) used by the PSO was designed to maximise both Matthews correlation coefficient MCC (Equation (4)) and α whilst, at the same time, minimising β . α and β were normalised between 0 and 1 when entered into the fitness function.

$$F = 2 \cdot \text{MCC} + \alpha - \beta \quad (3)$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(FP + FN)(TN + FP)(TN + FN)}} \quad (4)$$

MCC is a measure designed to evaluate the predictions of binary classifiers, which is especially robust in problems with high class imbalance, such as our dataset. MCC scores 1 for perfect classification, -1 for perfect misclassification, and 0 for equivalent to random classification. The F-measure is commonly used in information/document retrieval, as it does not fall prey to the same problems, however, it does not include true negatives in the calculation. This is not an issue within document retrieval, as they are not important to that particular task; however, in a detection system, correctly classifying negatives (true negatives) is an integral objective of the system.

It was desirable to attain a high α , as this corresponds to how far in advance the alert of oncoming disease is raised, whereas β needs to be minimised, as this represents the period of time the alert is for. For example, given the values $\alpha = 7$ and $\beta = 1$, we know that when an alert is raised, it is expected that occurrences of respiratory disease will appear within 7–8 days. Given a different detector for the values $\alpha = 3$ and $\beta = 5$, we know that when an alert is raised, it is expected that occurrences of respiratory disease will appear within 3–8 days, a larger window and, therefore, more uncertainty. As we consider MCC to be the most important component of our fitness function, it has a larger weight in Equation 3.

4.2. Other Metrics Used for Evaluation

Specificity, a commonly used evaluation metric, can produce misleading scores in datasets with a large number of negative examples (the typical scenario in anomaly detection tasks). Instead of specificity, we have used positive predictive value (also known as precision, Equation (5)). For the sake of brevity, we identify this metric as precision throughout this paper. We also made use of sensitivity (also known as recall, Equation (6)), which we identify as recall throughout this paper. We did not use

$$d = \sqrt{(e - \bar{R})^T \Sigma^{-1} (e - \bar{R})} \quad (2)$$

4.1.1. PSO优化异常检测

一旦所有帧的距离都计算完毕，下一步是使用PSO [37]优化阈值异常检测器，该检测器能够判断哪些帧应被视为异常。PSO [37]是一种元启发式算法，它利用群智能的概念来寻找一组超参数，从而得到近乎最优的解决方案。考虑一个序列中所有帧的马氏距离向量 $\vec{D} = \{d_0, d_1, d_2, \dots, d_n\}$ ，其中每个 d_i 是帧索引 i 处的重建误差与 R 之间的马氏距离。任何满足 $d_i : d_{i-1} < \tau \wedge d_i \geq \tau$ 的 d_i 被定义为警报。警报的有效性通过判断疾病在从 $i + \alpha$ 开始到 $\alpha + \beta$ 结束的可接受预测窗口内是否从0增加来评估。如果是，并且没有其他警报因疾病流行率的增加而触发，则该警报被视为真阳性，否则为假阳性。此外，考虑一个向量 $\vec{F} = \{f_0, f_1, f_2, \dots, f_n\}$ ，其中每个 f_i 包含帧 i 显示呼吸系统疾病症状的动物数量。任何满足 $i : f_{i-1} = 0 \wedge f_i > 0$ 且在 $i - \alpha$ 和 $i - \beta$ 之间没有警报的帧被确定为假阴性。

检测器的参数 τ 、 α 和 β 通过PSO进行优化。每个折独立地使用粒子进行优化 O_e ，然后对每个参数的最佳值在 O_t 上进行最终性能评估。每个粒子使用三个维度随机初始化，分别代表 τ 、 α 和 β 。PSO使用的适应度函数（公式 (3)）旨在最大化马修斯相关系数MCC（公式 (4)）和 α ，同时最小化 β 。 α 和 β 在输入适应度函数时被归一化到0和1之间。

$$F = 2 \cdot \text{MCC} + \alpha - \beta \quad (3)$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(FP + FN)(TN + FP)(TN + FN)}} \quad (4)$$

MCC是一个用于评估二元分类器预测的指标，在类别不平衡严重的问题中特别稳健，例如我们的数据集。MCC在完美分类时得分为1，在完美错误分类时得分为 -1 ，在随机分类时得分为0。F-measure在信息/文档检索中常用，因为它不会受到同样的问题影响，但它不包括真负例（true negatives）在计算中。在文档检索中这不是问题，因为它们对该特定任务不重要；但在检测系统中，正确分类负例（真负例）是系统的核心目标之一。

最好能达到高 α ，因为这对应于疾病预警提前的程度，而 β 需要最小化，因为这代表预警持续的时间。例如，给定值 $\alpha = 7$ 和 $\beta = 1$ ，我们知道当预警被触发时，预计呼吸系统疾病的发作将在7-8天内出现。对于不同的检测器，给定值 $\alpha = 3$ 和 $\beta = 5$ ，我们知道当预警被触发时，预计呼吸系统疾病的发作将在3-8天内出现，这是一个更大的窗口，因此不确定性也更大。由于我们认为MCC是我们适应度函数最重要的组成部分，它在公式 3. 中具有更大的权重。

4.2. 其他用于评估的指标

特异性，一个常用的评估指标，在包含大量负例（异常检测任务中的典型场景）的数据集中会产生误导性分数。我们未使用特异性，而是采用了阳性预测值（也称为精确率，公式 (5)）。为简洁起见，本文将此指标称为精确率。我们还使用了灵敏度（也称为召回率，公式 (6)），本文将其称为召回率。我们没有使用

precision or recall for the optimisation of the hyper-parameters of the detector. They were only used to evaluate the final performance of each method.

$$P = \frac{tp}{tp + fp} \tag{5}$$

$$R = \frac{tp}{tp + fn} \tag{6}$$

4.3. Methods Included for Comparison

All methods used were tested on identical test sets O_t . Once all methods for comparison were carried out, a Wilcoxon test was conducted to compare the precision, recall, and MCC between the three comparison methods and the GRU-AE. These results were corrected for multiple observations using Holm’s method [38].

4.3.1. Luminol

In order to provide a baseline for comparison of performance, LinkedIn’s open source, univariate time-series anomaly detection package, Luminol [39], was used. This library represents out-of-the-box anomaly detection, as well as methods which require data be univariate, often achieved using principal component analysis (PCA). This uses a bitmap representation of a univariate time series, calculated using symbolic aggregate approximation (SAX)—a method for the discretisation of time-series data [40]—to detect anomalies [41]. This is implemented using two concatenated windows, a lagged window and a leading window, which slide across the time series. Each window is then converted into a SAX representation and subsequently into bitmaps based on the frequency of each SAX subword. The distance between the two bitmaps is calculated and represents the anomaly score for the leading window. The calculated anomaly scores are then tested using a grid search-optimised threshold-based anomaly detector in the same way that the GRU-AE is assessed. In order to transform the three environmental variables into a univariate time series, the first principal component was taken.

4.3.2. Time-Series Regression with Autoregression Integrated Moving Average (ARIMA)

Autoregression integrated moving average (ARIMA) is a frequently used model for univariate time-series analysis in statistical analysis. The parameters for this model (p , the order of autoregressive model; d , the degree of differencing; and q , the order of the moving average model) were found using grid search on H_t , optimising for the lowest Akaike information criteria (AIC). This is a metric used for comparing the goodness of fit of statistical models, though it is not a measure of goodness of fit. Once the optimal parameters were found, H_o was used to fit the model. This was then used to forecast 30-minute windows, given the previous 180 minutes in O_e . Windows where the mean square error between the predicted window and ground truth was beyond some threshold (τ) were considered an alert for the window $i + \alpha$ to $i + \alpha + \beta$. Values for τ , α , and β were optimised using grid search and, finally, tested on O_t . Similar to the Luminol method, the first principal component was taken to transform the three environmental variables into a univariate time series.

4.3.3. Time-Series Regression with GRU Network (GRU-R)

To provide context for how the GRU-AE performs with regards to an alternative deep learning methodology, a GRU network was used for regression. This network was trained to predict future values of the environmental sensors. By training this regression network on only healthy data H_t , it learned to predict environmental variables for conditions which do not result in occurrences of respiratory disease. The model was trained to predict windows of 30 minutes, the same size as the windows used for the GRU-AE. Similar to the ARIMA model, windows where the mean square error was beyond some threshold (τ) were considered an alert for the window $i + \alpha$ to $i + \alpha + \beta$. Similar to

精确率或召回率来优化检测器的超参数。它们仅用于评估每种方法的最终性能。

$$P = \frac{tp}{tp + fp} \tag{5}$$

$$R = \frac{tp}{tp + fn} \tag{6}$$

4.3. 用于比较的方法

所有使用的方法都在相同的测试集上进行了测试 O_t 。一旦所有用于比较的方法都完成，就进行了Wilcoxon检验，以比较三种比较方法与GRU-AE之间的精确率、召回率和MCC。这些结果使用Holm方法进行了多重观察校正 [38]。

4.3.1. Luminol

为了提供性能比较的基线，使用了LinkedIn的开源、单变量时间序列异常检测包Luminol [39]，。该库代表了开箱即用的异常检测，以及需要数据为单变量的方法，通常通过主成分分析（PCA）实现。它使用单变量时间序列的位图表示，该表示使用符号聚合近似（SAX）计算——这是一种时间序列数据离散化方法 [40]——来检测异常 [41]。这是通过两个连接的窗口实现的，一个滞后窗口和一个领先窗口，它们在时间序列上滑动。每个窗口然后被转换为SAX表示，并随后根据每个SAX子词的频率转换为位图。计算两个位图之间的距离，并代表领先窗口的异常分数。计算出的异常分数然后使用与评估GRU-AE相同的方式使用网格搜索优化的基于阈值的异常检测器进行测试。为了将三个环境变量转换为单变量时间序列，取了第一个主成分。

4.3.2. 时间序列回归与自回归积分移动平均（ARIMA）

自回归积分移动平均（ARIMA）是统计分析中用于单变量时间序列分析的常用模型。该模型的参数（ p ，自回归模型的阶数； d ，差分阶数；以及 q ，移动平均模型的阶数）通过在 H_t 上进行网格搜索找到，以优化最低的赤池信息量标准（AIC）。这是一个用于比较统计模型拟合优度的指标，但它不是拟合优度的度量。找到最优参数后，使用 H_o 对模型进行拟合。然后使用它来预测30分钟窗口，给定前180分钟在 O_e 中的数据。预测窗口与真实值之间的均方误差超过某个阈值 (τ) 的窗口被视为对窗口 $i + \alpha$ 到 $i + \alpha + \beta$ 的警报。 τ 、 α 和 β 的值通过网格搜索优化，最后在 O_t 上进行测试。与Luminol 方法类似，取第一主成分将三个环境变量转换为单变量时间序列。

4.3.3. T 使用GRU网络的时间序列回归 (GRU-R)

为了提供关于GRU-AE与另一种深度学习方法相比表现如何的背景，使用了一个GRU网络进行回归。该网络被训练以预测环境传感器未来的值。通过仅使用健康数据 H_t 训练这个回归网络，它学会了预测不会导致呼吸系统疾病发生的条件下的环境变量。该模型被训练以预测30分钟窗口，与GRU-AE使用的窗口大小相同。类似于ARIMA模型，均方误差超过某个阈值 (τ) 的窗口被视为对窗口 $i + \alpha$ 到 $i + \alpha + \beta$ 的警报。类似于

the configuration laid out in Section 4.1, these parameters were optimised using PSO, which is trained using O_e , and finally evaluated on O_t .

5. Results

For the purpose of exploration, the GRU-AE’s anomaly detector was initially optimised using a grid search with a fixed $\beta = 6$ with O_v used as the test set. For values from $\alpha = 10$ onwards, there was a steady and sustained decrease in the total number of alerts raised (Figure 5). This was indicative of the point where the acceptable forecast window extended beyond the data available, resulting in an increased portion of alerts that were unverifiable in terms of validity. This decline in alerts signified the maximum range of the acceptable forecast window $\alpha + \beta = 16$. This maximum endpoint was used to reduce the search space of the PSO, and, therefore, a particle was considered invalid if $\alpha + \beta > 16$. The search space was also restricted by stipulating that a particle was also invalid if it exceeded the following ranges: $0 \leq \tau \leq 20$, $1 \leq \alpha \leq 12$, and $1 \leq \beta \leq 6$. The limits on τ cover all feasible values; this was found through preliminary testing. β was limited to a maximum of 6, as a range greater than this was deemed to be too large.

As outlined in Section 4.1, grid search was used to find the hyper-parameters for the GRU-AE. The grid search converged to the same set of parameters in almost all batches, so, for the sake of space, we report in Table 1 the most frequently identified values for all hyper-parameters. As expected, the hidden size was very low. This is because under healthy circumstances there is little fluctuation in the environment, and, therefore, only a small amount of memory is required. It is when there are deviations from this—frequent or drastic fluctuations—that occurrences of respiratory disease can be expected.

Table 1. The grid search-optimised hyper-parameters used to train a GRU-AE used on multidimensional time-series data.

Hyper-Parameter	Value
Hidden size	4
No of layers	20
Mini-batch size	1024
Dropout	0.5
Learning rate	1×10^{-6}
Momentum	0.3

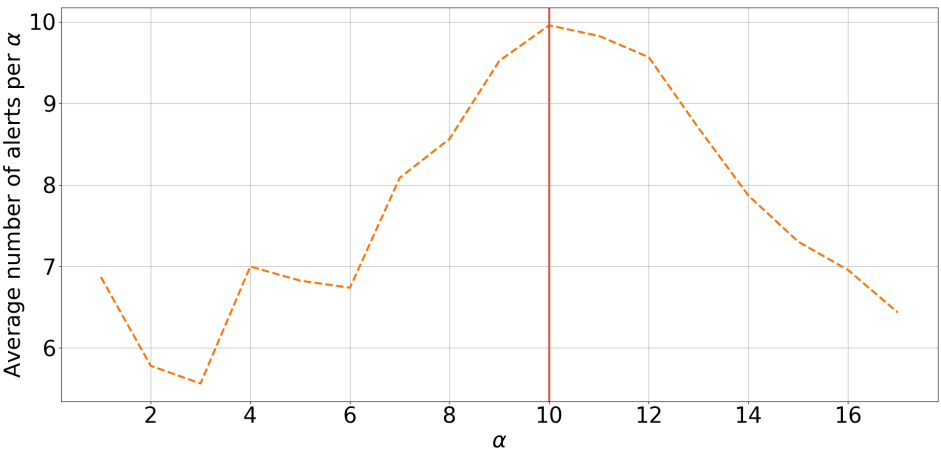


Figure 5. The average number of alerts raised, given an alert window start of α days and a size of $\beta = 7$, using the optimal threshold τ for each given α found using grid search on the anomaly detection used on Mahalanobis distances produced by a GRU-based autoencoder.

第4.1节中概述的配置，这些参数使用粒子群优化算法 进行了优化，该算法使用 O_e 进行训练，并最终在 O_t 上进行评估。

5. 结果

为探索目的，GRU-AE的异常检测器最初使用网格搜索进行了优化，其中固定使用 O_v 作为测试集。从 $\alpha = 10$ 开始，警报总数呈稳定且持续的下降（图5）。这表明可接受的预测窗口已超出可用数据范围，导致部分警报在有效性方面无法验证。警报的减少表明可接受的预测窗口的最大范围 $\alpha + \beta = 16$ 。该最大端点用于缩小PSO的搜索空间，因此，如果 $\alpha + \beta > 16$ ，则粒子被视为无效。搜索空间还通过规定如果粒子超出以下范围则无效来限制： $0 \leq \tau \leq 20$ 、 $1 \leq \alpha \leq 12$ 和 $1 \leq \beta \leq 6$ 。对 τ 的限制涵盖了所有可行值；这是通过初步测试发现的。 β 限制为最大6，因为大于此范围的值被认为太大。

如第 4.1节所述，采用网格搜索方法来确定GRU-AE的超参数。网格搜索在几乎所有批次中都收敛到相同的参数集，因此，为了节省篇幅，我们在表 1 中报告了所有超参数出现频率最高的值。正如预期，隐藏层大小非常小。这是因为健康状态下环境变化很小，因此只需要少量内存。只有当出现偏离这种情况时——频繁或剧烈的变化——才会出现呼吸系统疾病。

表1。 用于训练多维时间序列数据的GRU-AE的网格搜索优化超参数。

超参数	值
隐藏层大小	4
层数	20
小批量大小	1024
dropout	0.5
学习率	1×10^{-6}
动量	0.3

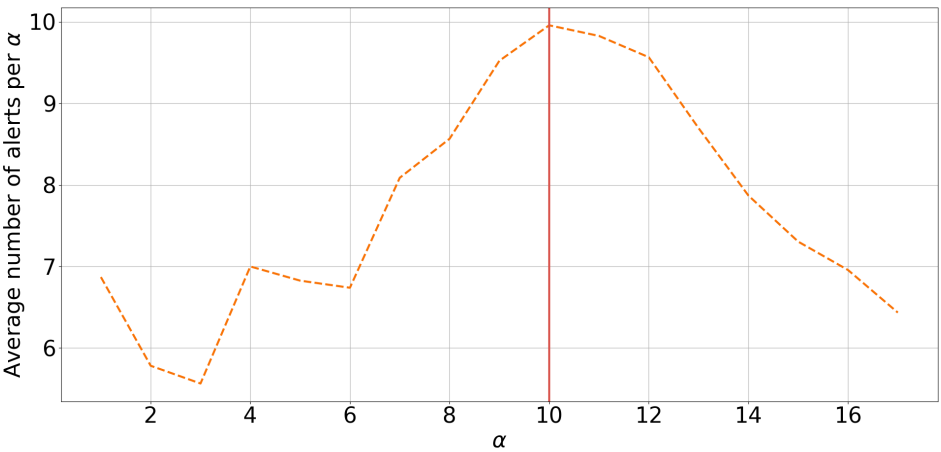


图5。 给定警报窗口起始时间为 α 天和大小为 $\beta = 7$ 时，平均触发的警报数量，使用通过网格搜索在马氏距离异常检测上找到的最优阈值 τ （针对每个给定的 α ），该异常检测由基于GRU的自动编码器产生。

Table 2 shows the batch-by-batch performance of each of the three methodologies; each row presents the performance of the three compared systems when a given batch is used for testing. The GRU-AE achieves better results than the other methods for all performance metrics measured (precision, recall, and MCC). The most notable difference between the proposed GRU-AE and other methodologies is the substantial shift in performance in terms of precision and recall; ARIMA has also achieved this, but to a much lower degree. In the Luminol and GRU-R models, a higher precision is achieved in comparison to recall, indicating that these models are more conservative in their alerts. However, in the GRU-AE model, the solutions found tend to rely on maximising recall at the expense of precision, $\mu_P = 0.776$, $\mu_R = 0.938$, whilst not sacrificing it so much to be degrading to the model’s overall performance. The substantially higher recall for the GRU-AE model indicates the models capacity to identify negative changes in the environment that lead to an increased number of pigs with symptoms of respiratory disease; the lower precision of the model could indicate that not all negative changes in the environment result in occurrences of respiratory disease. This is backed up by the fact that there are other factors [42] that contribute to respiratory disease prevalence, not just the environmental measures used in this analysis.

The last row of Table 2 contains a high-level quantitative summary of the performance of all methods across batches. We have counted the number of batches in which each method obtains either an incorrect score of 0 for MCC (I), correct score of 1 for MCC (C), or in between 0 and 1 (B). In terms of the C count, Luminol clearly performs the worst of the methods, while the others have a similar score (10–11). At the same time, GRU-AE stands out as having the lowest count of I, twice as low as the score of ARIMA and GRU-R, and almost four times lower than Luminol’s, whilst simultaneously keeping the lowest count of B.

Numerically, at least, we can see that the difference in performance between Luminol and GRU-AE may be considered substantial. However, the results from the Wilcoxon test showed that the difference was not statistically significant for precision ($p = 0.184$), recall ($p = 0.059$), and MCC ($p = 0.054$). The Wilcoxon test also concluded there was no significant difference between the ARIMA model and GRU-AE, and GRU-R and GRU-AE models, in terms of any of the performance metrics.

Looking only at test batches where there were no events (Table 2), the deep learning models excel in determining that there are no alerts to be raised, achieving extremely high precision and recall. This indicates that the GRU is capable of modelling the underlying representation of “normality” very well, as alerts are not raised when they should not be. For batches where the number of events was greater than zero, the ARIMA model outperformed both of the GRU-based models in terms of precision. However, this came at a substantial sacrifice in terms of recall, indicating that the model was likely underpowered. Both of the GRU-based models decreased in performance in terms of precision, but only the GRU-AE was able to retain its high recall. The optimised α and β values for each batch are reported in Table 3.

表 2展示了三种方法的批次性能；每一行展示了当使用某个批次进行测试时，三个比较系统的性能。GRU-AE在所有性能指标（精确率、召回率和MCC）上都优于其他方法。GRU-AE与其他方法最显著的区别在于精确率和召回率上的性能大幅提升；ARIMA也实现了这一点，但程度要低得多。在Luminol和GRU-R模型中，精确率高于召回率，表明这些模型在警报方面更为保守。然而，在GRU-AE模型中，找到的解决方案倾向于以牺牲精确率为代价来最大化召回率， $\mu_P = 0.776$ $\mu_R = 0.938$ ，但并未严重牺牲模型的总体性能。GRU-AE模型显著更高的召回率表明其能够识别导致猪出现呼吸系统疾病症状的环境负面变化；模型的精确率较低可能表明并非所有环境负面变化都会导致呼吸系统疾病的发生。这一点有事实支持，即除了本次分析中使用的环境测量指标外，还有其他因素 [42] 导致呼吸系统疾病的发生。

表2的最后行包含了所有方法在各个批次中的性能高级定量总结。我们统计了每种方法获得MCC（MCC）错误分数0（I）、正确分数1（C）或介于0和1之间（B）的批次数量。在C计数方面，Luminol明显是所有方法中表现最差的，而其他方法得分相似（10–11）。同时，GRU-AE的I计数最低，是ARIMA和GRU-R得分的两倍，几乎是Luminol的四倍，同时B计数也最低。

从数值上看，我们至少可以看到Luminol和GRU-AE之间的性能差异可能很大。然而，Wilcoxon检验的结果显示，在精确率（ $p = 0.184$ ）、召回率（ $p = 0.059$ ）和MCC（ $p = 0.054$ ）方面，差异没有统计学意义。Wilcoxon检验还得出结论，ARIMA模型和GRU-AE模型、GRU-R和GRU-AE模型在性能指标方面没有显著差异。

仅观察无事件（表2）的测试批次，深度学习模型在判断无需发出警报方面表现出色，实现了极高的精确率和召回率。这表明GRU能够很好地建模“正常”状态的基础表示，因为当不应发出警报时，模型确实不会发出警报。对于事件数量大于零的批次，ARIMA模型在精确率方面优于两种基于GRU的模型。然而，这是以召回率大幅牺牲为代价的，表明模型可能能力不足。两种基于GRU的模型在精确率方面性能均有所下降，但只有GRU-AE能够保持其高召回率。每个批次的优化 α 和 β 值见表3。

Table 2. Table of results for the grid search-optimised implementation of LinkedIn’s anomaly detection library (Luminol), the grid search-optimised autoregression integrated moving average (ARIMA) model, particle swarm optimisation (PSO)-optimised threshold-based anomaly detection of loss incurred from a GRU-based regression, and PSO-optimised threshold-based anomaly detection of Mahalanobis distance produced by a GRU-based autoencoder. The events column lists how many times the respiratory disease prevalence increases from 0 within the batch, P denotes the precision, and R is the recall of the model. These results are from the test data O_t . The final I/B/C row indicates the number of folds that were Incorrect ($MCC = 0.0$), Between ($0.0 < MCC < 1.0$), or Correct ($MCC = 1.0$)

ID	Events	Luminol			ARIMA			GRU-R			GRU-AE		
		P	R	MCC	P	R	MCC	P	R	MCC	P	R	MCC
5	0	0.111	0.125	0.118	1.0	1.0	1.0	1.0	1.0	1.0	0.0	0.0	0.0
9	0	1.0	0.1	0.316	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
10	0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	1.0
14	0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
16	0	1.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
17	0	1.0	0.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
18	0	1.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
19	0	1.0	0.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
24	0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
26	4	1.0	0.667	0.816	0.053	0.5	0.162	1.0	0.0	0.0	0.250	1.0	0.500
27	5	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.111	1.0	0.333
28	3	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	1.0
29	1	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	0.0
30	0	0.0	0.0	0.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
31	1	0.0	0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	0.050	1.0	0.224
32	0	0.0	0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Avg. Events = 0		0.647	0.505	0.551	0.727	1.000	0.727	0.909	0.909	0.909	0.909	0.909	0.909
Avg. Events > 0		0.200	0.133	0.163	0.811	0.500	0.432	0.400	0.400	0.200	0.482	1.000	0.411
Avg. All		0.507	0.390	0.430	0.753	0.844	0.635	0.750	0.688	0.688	0.776	0.938	0.754
I/B/C		7/5/4			5/1/10			5/0/11			2/3/11		

表2。表为网格搜索优化的领英异常检测库（Luminol）、网格搜索优化的自回归集成移动平均（ARIMA）模型、基于GRU的回归损失 incurredfrom 的PSO优化阈值异常检测，以及基于GRU自动编码器产生的马氏距离的PSO优化阈值异常检测的结果。事件列显示呼吸系统疾病在批次中从0增加的次数，P表示精确率，R是模型的召回率。这些结果来自测试数据 O_{t_0} 。最终的I/B/C行表示错误（ $MCC = 0.0$ ）、介于（ $0.0 < MCC < 1.0$ ）或正确（ $MCC = 1.0$ ）的折数。

ID	事件	Luminol			ARIMA			GRU-R			GRU-AE		
		P	R	MCC	P	R	MCC	P	R	MCC	P	R	MCC
5	0	0.111	0.125	0.118	1.0	1.0	1.0	1.0	1.0	1.0	0.0	0.0	0.0
9	0	1.0	0.1	0.316	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
10	0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	1.0
14	0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
16	0	1.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
17	0	1.0	0.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
18	0	1.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
19	0	1.0	0.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
24	0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
26	4	1.0	0.667	0.816	0.053	0.5	0.162	1.0	0.0	0.0	0.250	1.0	0.500
27	5	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.111	1.0	0.333
28	3	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	1.0
29	1	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	0.0
30	0	0.0	0.0	0.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
31	1	0.0	0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	0.050	1.0	0.224
32	0	0.0	0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
平均事件 = 0		0.647	0.505	0.551	0.727	1.000	0.727	0.909	0.909	0.909	0.909	0.909	0.909
平均事件 > 0		0.200	0.133	0.163	0.811	0.500	0.432	0.400	0.400	0.200	0.482	1.000	0.411
平均所有		0.507	0.390	0.430	0.753	0.844	0.635	0.750	0.688	0.688	0.776	0.938	0.754
I/B/C		7/5/4			5/1/10			5/0/11			2/3/11		

Table 3. Table of results for the grid search-optimised implementation of LinkedIn’s anomaly detection library (Luminol), the grid search-optimised ARIMA model, PSO-optimised threshold-based anomaly detection of loss incurred from a GRU-based regression, and PSO-optimised threshold-based anomaly detection of the Mahalanobis distance produced by a GRU-based autoencoder. α and β denote the start and length of the time window (in days), respectively, for which an alert is assessed, and τ is the threshold the loss/Mahalanobis distance needed to exceed in order to raise an alert. They are optimised using training data O_e ; these results are from the test data O_t .

ID	Events	Luminol		ARIMA		GRU-R		GRU-AE	
		α	β	α	β	α	β	α	β
5	0	12	4	12	1	1	6	1	1
9	0	12	4	12	1	1	6	1	2
10	0	12	4	12	1	1	6	1	2
14	0	12	4	12	1	1	6	1	4
16	0	12	4	12	1	1	6	3	1
17	0	12	4	12	1	1	6	1	5
18	0	12	4	12	1	1	6	3	5
19	0	12	4	12	1	1	6	1	5
24	0	12	4	12	1	1	6	1	6
26	4	12	4	12	1	2	6	3	6
27	5	12	4	12	4	4	6	1	6
28	3	12	4	12	1	5	6	8	6
29	1	12	4	12	2	1	6	2	6
30	0	12	4	12	1	1	6	4	6
31	1	12	4	12	1	1	6	1	6
32	0	12	4	12	1	1	6	8	5

5.1. Case Study

The following case is presented for batch 27 exemplifies the characteristics of each of the methodologies applied to the problem. This batch was chosen as it contained the highest number of anomalous events in a single batch, making it one of the more challenging batches for a method to perform well on. The batch was analysed by a Luminol-powered, grid search-optimised anomaly detection (Figure 6), an ARIMA-powered, grid search-optimised anomaly detection (Figure 7, a GRU-R-powered, PSO-optimised anomaly detection (Figure 8), and a GRU-AE-powered, PSO-optimised anomaly detection (Figure 9). The first subplot for each figure shows respiratory disease prevalence in both analogue and binary form; the second plots the univariate time series that was used for anomaly detection and the alerts raised for it. In the case of the Luminol model, this was the first principal component of the temperature, humidity, and CO₂ time series. For the GRU-R and ARIMA models, the mean square error between predicted and ground truth was used; for the GRU-autoencoder, the binary cross-entropy loss between each input window and the model’s recreation of it was used. The final subplot shows the results of the anomaly detection for each of the alerts, along with their alert windows.

The two deep learning-based models show signs of consistency in the underlying technology, as they both capture similar events within the environmental data, demonstrated by the common peaks and troughs seen in the loss. The Luminol-based model correctly produces alerts to some of the occurrences of respiratory disease prevalence, but these are severely outweighed by the number of false positives that it also produces. The GRU-R and ARIMA models suffer from the inverse problem in that no alerts are raised, even though the latter appears at first glance to produce some significant anomalous points. These points rarely align with actual events, and the success of the ARIMA model is largely down to not raising many alerts at all. The GRU-autoencoder model, however, strikes a balance between these solutions, alerting to almost all of the occurrences of respiratory disease, albeit with some false positives, though far fewer than the Luminol model.

表3。 表为网格搜索优化的领英异常检测库（Luminol）、网格搜索优化的ARIMA模型、基于GRU的回归损失PSO优化阈值异常检测以及基于GRU自动编码器马氏距离PSO优化阈值异常检测的结果。 α 和 β 分别表示评估警报的时间窗口的起始和长度（以天为单位）， τ 是需要超过的损失/马氏距离阈值以触发警报。它们使用训练数据 O_e 进行优化；这些结果来自测试数据 O_t 。

ID	事件	Luminol		ARIMA		GRU-R		GRU-AE	
		α	β	α	β	α	β	α	β
5	0	12	4	12	1	1	6	1	1
9	0	12	4	12	1	1	6	1	2
10	0	12	4	12	1	1	6	1	2
14	0	12	4	12	1	1	6	1	4
16	0	12	4	12	1	1	6	3	1
17	0	12	4	12	1	1	6	1	5
18	0	12	4	12	1	1	6	3	5
19	0	12	4	12	1	1	6	1	5
24	0	12	4	12	1	1	6	1	6
26	4	12	4	12	1	2	6	3	6
27	5	12	4	12	4	4	6	1	6
28	3	12	4	12	1	5	6	8	6
29	1	12	4	12	2	1	6	2	6
30	0	12	4	12	1	1	6	4	6
31	1	12	4	12	1	1	6	1	6
32	0	12	4	12	1	1	6	8	5

5.1. 案例研究

以下案例以第27批次为例，说明了应用于该问题的每种方法的特点。该批次被选中，因为它包含单个批次中异常事件数量最多，是方法表现较好的更具挑战性的批次之一。该批次通过Luminol驱动、网格搜索优化的异常检测（图6）进行分析，ARIMA驱动、网格搜索优化的异常检测（图7），GRU-R驱动、粒子群优化算法优化的异常检测（图8），以及GRU-AE驱动、粒子群优化算法优化的异常检测（图9）。每个图的第一个子图显示了呼吸系统疾病在模拟和二进制形式中的流行情况；第二个子图绘制了用于异常检测的单变量时间序列及其产生的警报。对于Luminol模型，这是温度、湿度和CO₂ 时间序列的第一主成分。对于GRU-R和ARIMA模型，使用预测值与真实值之间的均方误差；对于GRU自动编码器，使用每个输入窗口与模型重建之间的二元交叉熵损失。最后一个子图显示了每个警报的异常检测结果及其警报窗口。

这两种基于深度学习的模型在底层技术方面显示出一致性，因为它们都能在环境数据中捕捉到相似的事件，这在损失曲线中的共同峰值和谷值中得到了证明。基于Luminol的模型能够正确地有部分呼吸系统疾病高发情况进行警报，但这些警报被其产生的大量误报远远超过了。而GRU-R和ARIMA模型则存在相反的问题，即即使后者在初看时似乎产生了一些显著的异常点，但也没有发出任何警报。这些点很少与实际事件相符，ARIMA模型的成功很大程度上归功于它几乎不发出任何警报。然而，GRU自动编码器模型在这两种解决方案之间取得了平衡，它对几乎所有呼吸系统疾病的发生情况都发出了警报，尽管存在一些误报，但数量远少于基于Luminol的模型。

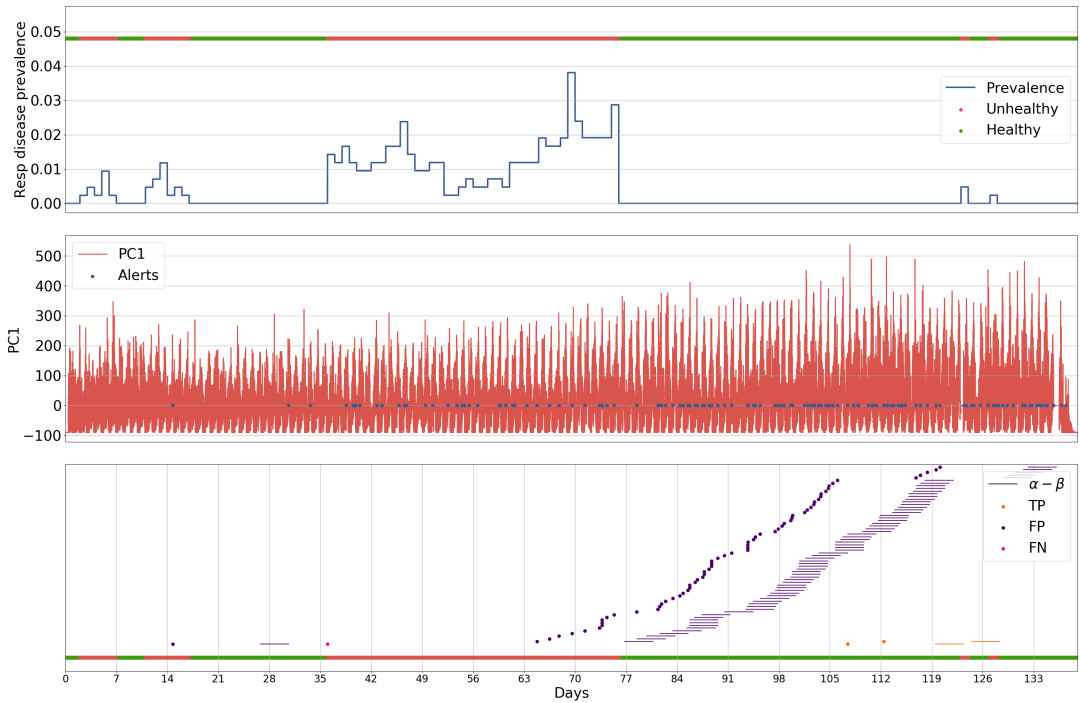


Figure 6. Results for batch 27 processed by the Luminol-powered anomaly detection, given its grid search-optimised parameters α (the start of the window for which an alert is for) and β (the length of the window), in relation to the prevalence of respiratory disease within a single batch.

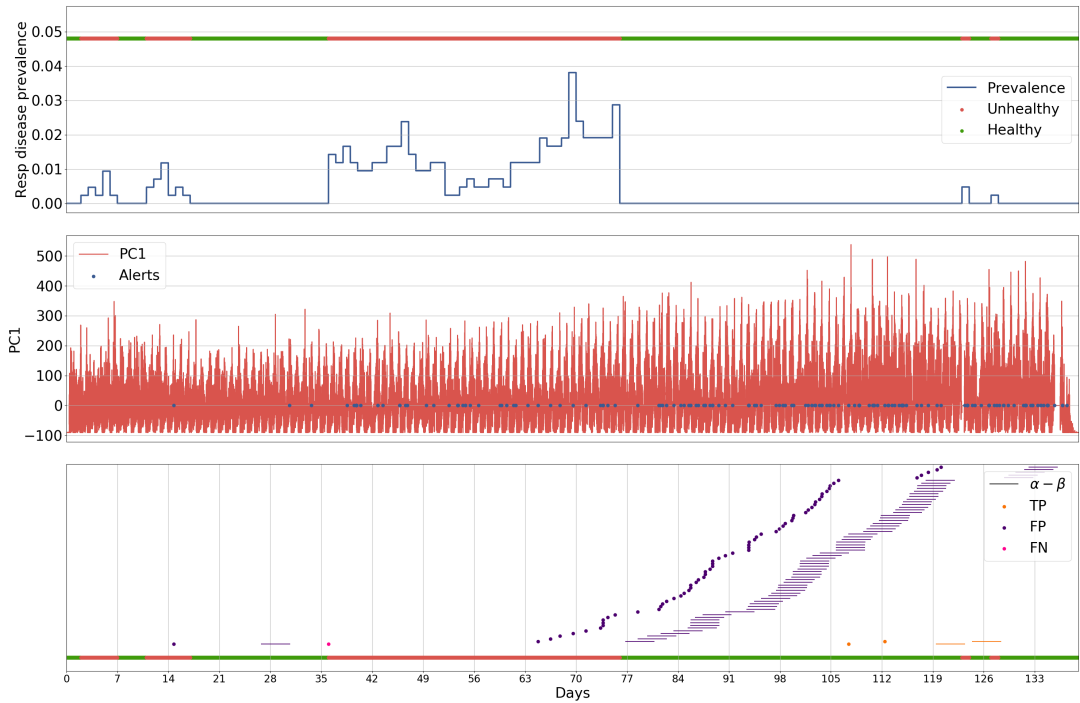


图6. 使用Luminol驱动异常检测处理批次27的结果，考虑了其网格搜索优化的参数 α （警报窗口的起始位置）和 β （窗口长度），与单个批次中呼吸系统疾病的流行率之间的关系。



Figure 7. Results for batch 27 processed by the ARIMA-powered anomaly detection, given its grid search-optimised parameters α (the start of the window for which an alert is for), β (the length of the window), and τ (the threshold which the loss between predicted values and actual values must cross to be considered an alert), in relation to the prevalence of respiratory disease within a single batch.



图7. 使用ARIMA驱动异常检测处理批次27的结果，考虑了其网格搜索优化的参数 α （警报窗口的起始位置）、 β （窗口长度）和 τ （预测值与实际值之间的损失必须跨越的阈值，才能被视为警报），与单个批次中呼吸系统疾病的流行率之间的关系。



Figure 8. Results for batch 27 processed by the GRU-R-powered anomaly detection, given its PSO-optimised parameters α (the start of the window for which an alert is for), β (the length of the window), and τ (the threshold which the loss between predicted values and actual values must cross to be considered an alert), in relation to the prevalence of respiratory disease within a single batch.



图8. 使用GRU-R驱动异常检测（基于PSO优化参数 α （警报窗口的起始位置）、 β （窗口长度）和 τ （预测值与实际值之间的损失必须跨越的阈值以被视为警报）处理批次27的结果，与单个批次中呼吸系统疾病的流行率的关系。

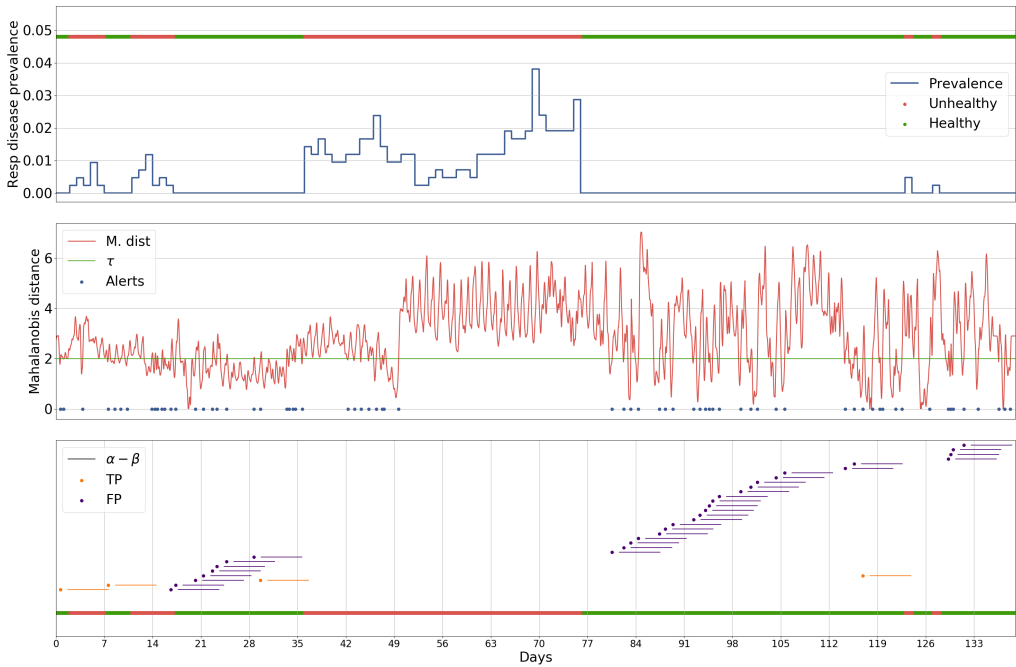


Figure 9. Results for batch 27 processed by the GRU-autoencoder-powered anomaly detection, given its PSO-optimised parameters α (the start of the window for which an alert is for), β (the length of the window), and τ (the threshold which the loss between predicted values and actual values must cross to be considered an alert), in relation to the prevalence of respiratory disease within a single batch.

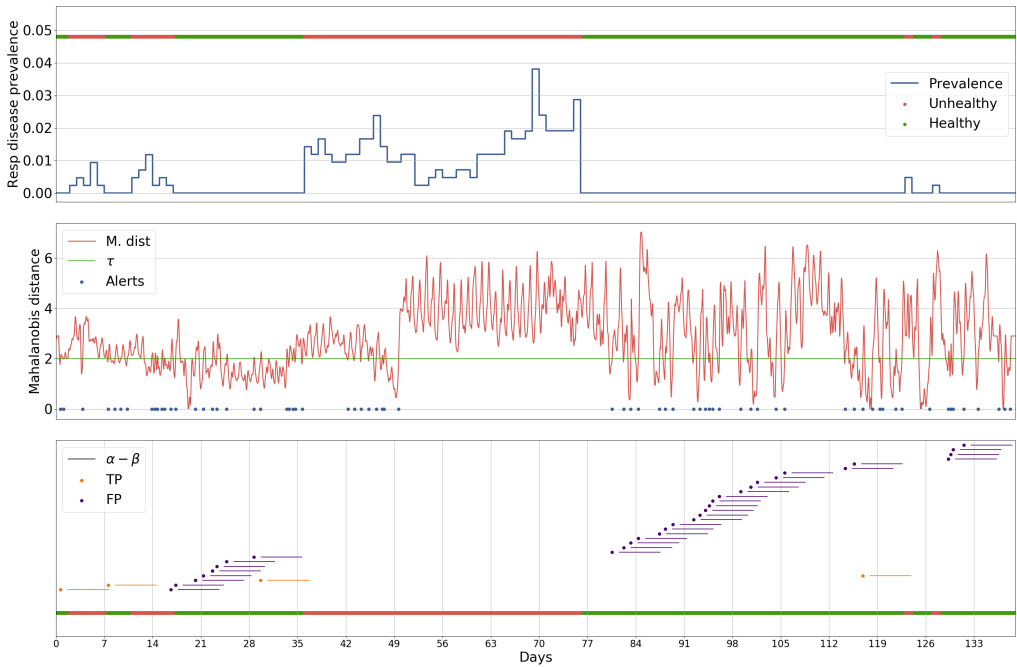


图9. 使用GRU自动编码器驱动异常检测（基于PSO优化参数 α （警报窗口的起始位置）、 β （窗口长度）和 τ （预测值与实际值之间的损失必须跨越的阈值以被视为警报）处理批次27的结果，与单个批次中呼吸系统疾病的流行率的关系。

5.2. Influence of the Number of Hidden Layers on GRU-AE Performance

The number of layers used in the encoder and decoder had an effect on what the GRU-AE was capable of modelling (Figure 10). Since the concept behind the model was that loss is reduced for “healthy” data, the perfect environment’s loss would have a constant loss over time, therefore meaning the model architecture should be chosen to minimise the standard deviation of the loss in healthy data. Using a small number of layers, 1–15, there was little change in the ability of the GRU-AE to model “healthy data”. Once the number of layers reached 20 for each half of the GRU-AE, the performance reached its maximum and would not reduce the standard deviation of loss any further.

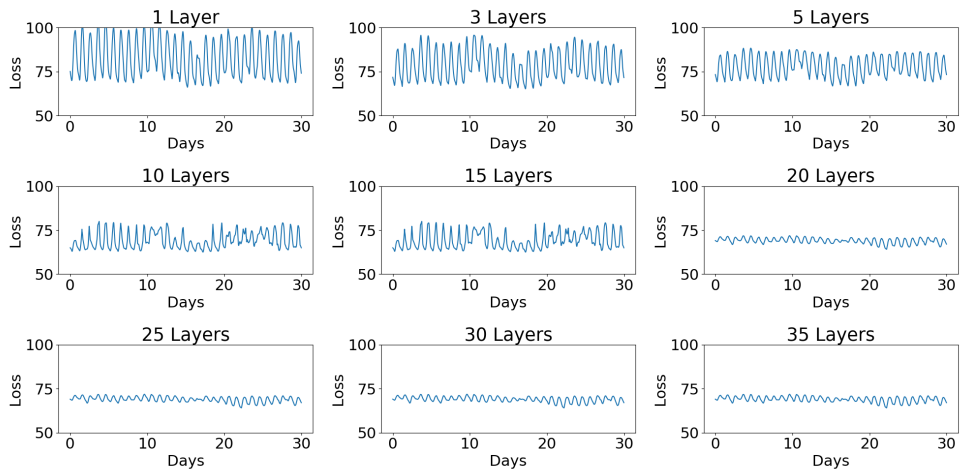


Figure 10. How different numbers of layers in a GRU-AE affect the ability to model environmental data. The smaller the standard deviation, the better the model performs.

5.3. Computation Time

Despite its lacking in performance, the Luminol-based model presented (Figure 6) was trained and optimised in a fraction of the time required to train the two GRU models presented. The Luminol model was fully trained within a period of 6 h and the ARIMA-based model was fully trained within 24 h. These models were run on a machine containing two Xeon E5-2699v4 processors (totalling 44 cores), whereas the two GRU models took a week on a similarly performing machine equipped with an Nvidia GTX 980 Ti.

6. Discussion

This paper set out to develop and evaluate a methodology utilising recurrent neural networks comprising GRUs, and to apply the methodology to the early detection of respiratory disease in growing pigs. This network structure was used to build an autoencoder capable of accounting for the temporal aspect of data within the structure of the network itself, rather than solely through the structuring of the data. The GRU-AE specifically outperforms the GRU-R where there were events to be detected, resulting in a substantially higher recall for the GRU-AE.

The novelty of this work lies in the complete reassessment of the idea of a recurrent autoencoder for multi-sensor anomaly detection of Malhotra et al. [3]. The primary adjustments made were switching from LSTM cells, currently the most common unit in recurrent networks, to GRUs. This benefits the network as a whole, as GRUs are less complex and therefore more efficient whilst providing comparable performance [43–46]. A PSO-optimised anomaly detector was also created for processing the output of the GRU-AE to determine if a window was anomalous or not. In addition to the methodological changes, this work provides a much more in-depth evaluation of the performance

5.2. 门控循环单元-自动编码器层数对性能的影响

编码器和解码器中使用的层数对GRU-AE能够建模的内容有影响（图10）。由于模型背后的概念是“健康”数据的损失会减少，因此完美环境的损失会随时间保持恒定，这意味着模型架构应该选择以最小化健康数据中的损失标准差。使用较少的层数，1-15层，GRU-AE建模“健康数据”的能力几乎没有变化。一旦GRU-AE每半部分的层数达到20层，性能达到最大值，并且不会进一步降低损失的标准差。

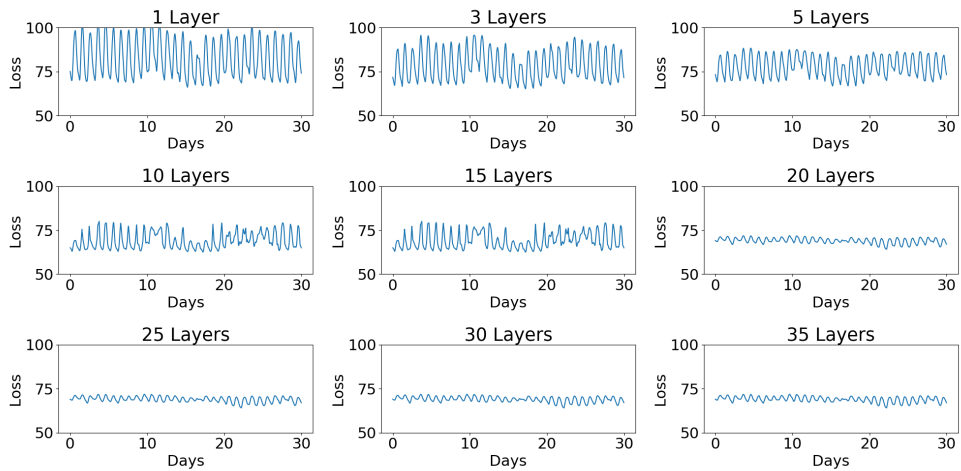


图10. GRU-AE中不同层数如何影响建模环境的能力数据。标准差越小，模型性能越好。

5.3. 计算时间

尽管性能有所欠缺，所提出的基于Luminol的模型（图6）在训练和优化上所需时间仅为所提出的两个门控循环单元模型所需时间的一小部分。基于Luminol的模型在6小时内完成完全训练，而基于ARIMA的模型在24小时内完成完全训练。这些模型在一台配备两个Xeon E5-2699v4处理器（总共44个核心）的机器上运行，而两个门控循环单元模型则在一台性能相似、配备Nvidia GTX 980 Ti的机器上花费了一周时间。

6. 讨论

本文旨在开发并评估一种利用包含门控循环单元的循环神经网络的方法，并将该方法应用于生长猪呼吸系统疾病的早期检测。该网络结构被用于构建一个自编码器，该自编码器能够在网络结构本身中考虑数据的时序性，而不仅仅是通过数据的结构化。GRU-AE在存在需要检测的事件时，具体表现优于GRU-R，从而为GRU-AE带来了显著更高的召回率。

这项工作的创新之处在于对 Malhotra 等人提出的用于多传感器异常检测的循环自编码器想法进行了全面重新评估。主要调整是将目前循环网络中最常用的 LSTM 单元切换为 GRU。这对整个网络都有益，因为 GRU 更简单，因此更高效，同时性能相当 [43–46]。此外，还创建了一个 PSO 优化异常检测器，用于处理 GRU-AE 的输出，以确定一个窗口是否异常。除了方法论上的变化，这项工作还通过案例研究和与其他替代方法的实际比较，对 GRU-AE 的性能进行了更深入的评估，并将该方法应用于真实世界数据。

of the GRU-AE through the use of a case study and concrete comparisons with alternative methods, whilst also applying the methodology to real-world data.

Given the recent trends in livestock keeping, there is a move towards automating significant portions of livestock management and monitoring in response to the ever-increasing demand for high-quality, sustainable, and safe meat production [47]. These changes in livestock management offer opportunities for automation in disease detection, such as the one developed here. A common approach to embracing what is generally referred to as precision livestock farming (PLF) is the introduction of a variety of sensors. Small, relatively cheap, and high-resolution sensors can monitor environmental variables, such as those assessed in this study, and also capture large quantities of other data using sensors such as cameras [48] and microphones [49]. In the case of disease detection, any false negatives have the potential to lead to epidemics and therefore increase the disease risk of livestock systems. For this reason, a high precision ought to be sacrificed if a higher recall can be achieved, as false positives pose less danger than false negatives in this application. The GRU-AE presented in this paper acts upon this, ensuring that recall is maximised as much as possible.

The motivation for developing such a methodology was to deal with the very low disease prevalence in the data that was collected. In this work, we concentrated on the detection of respiratory diseases. There were several reasons for this, including the fact that the prevalence of respiratory diseases was highest amongst this class of pigs in this dataset. We have similar data for other disorders that affect pigs, such as lameness and digestive diseases, but as their incidence is much lower than respiratory diseases, developing similar methodologies would result in an insufficient number of occurrences to validate the model.

Steps were taken to mitigate the challenges posed by the data, such as summing all classifications of respiratory disease (light, mild, severe, and irrecoverable), as described in Section 3.2. This was justified, as this methodology is for an early-warning system, which only needs to know if there is disease or if there is not. Therefore, looking at respiratory disease prevalence as a whole, rather than segmented into levels of severity, makes for a simpler dataset that is more focussed on the challenge being tackled. It should be noted that the methodology developed here can be applied to all types of respiratory diseases in pigs.

The detection of respiratory disease following a change in the environmental condition fits within the parameters derived from literature, such that we are not predicting oncoming disease before pathogen incubation (e.g., PRRS has an average incubation time of 14 days [32]). Both the GRU-R and GRU-AE are able to raise an alert for oncoming respiratory disease 1–7 days prior to pigs in a batch first showing symptoms. The Luminol model does not reflect this, however, this is likely due to its inability to find hyper-parameters that result in a strong MCC. This causes the PSO to maximise the values for α and β , as there is no repercussion for doing so if the performance is poor regardless.

Due to the renewed popularity of RNN-based models for sequence analysis, a result of their exceptional performance in fields such as natural language processing, there is a need for methods to increase the interpretability of these black-box models. Work in this area has steadily grown [50,51], and future work might consider applying such methods to the model to gain a better understanding of the risk factors for respiratory disease in growing pigs based on the GRU-AE. As this method is assumption-free and works without considering differences between countries, additional work to validate that this multi-sensor anomaly detection scales beyond the domain of health within agriculture, thus validating its robustness, would be valuable to the wider community.

7. Conclusions

We have presented a deep learning-based model capable of raising an early warning to occurrences of respiratory disease in growing pigs. The proposed methodology of a GRU-based autoencoder combined with a PSO-optimised anomaly detector shows strong potential in its ability to detect anomalies that will lead to occurrences of respiratory disease in growing pigs, and it is robust to country-specific environments. This GRU-AE was able to outperform other comparable methodologies

通过案例研究和与其他替代方法的实际比较, 对 GRU-AE 的性能进行了更深入的评估, 并将该方法应用于真实世界数据。

鉴于畜牧业最近的趋势, 人们正朝着自动化畜牧业管理和监测的重大部分迈进, 以应对对高质量、可持续和安全肉类生产日益增长的需求 [47]。这些畜牧业管理的变化为疾病检测自动化提供了机会, 例如这里开发的方法。一种普遍采用的方法, 通常被称为精准畜牧业 farming (PLF), 是引入各种传感器。小型、相对便宜、高分辨率的传感器可以监测环境变量, 例如本研究中评估的变量, 并使用摄像头等传感器捕获大量其他数据 [48] 和麦克风 [49]。在疾病检测的情况下, 任何假阴性都有可能流病, 因此会增加畜牧业系统的疾病风险。出于这个原因, 如果可以实现更高的召回率, 就应该牺牲精确率, 因为在这种情况下, 假阳性带来的危险小于假阴性。本文提出的 GRU-AE 正是作用于这一点, 确保尽可能最大化召回率。

开发这种方法的动机是处理收集到的数据中疾病患病率非常低的问题。在这项工作中, 我们专注于呼吸系统疾病的检测。有几个原因促使我们这样做, 其中包括该数据集中这类猪的呼吸系统疾病患病率最高。我们也有其他影响猪的疾病的数据, 如跛行和消化系统疾病, 但由于它们的发病率远低于呼吸系统疾病, 开发类似的方法会导致模型验证的样本数量不足。

为应对数据带来的挑战, 我们采取了措施, 例如将所有呼吸系统疾病的分类(轻微、轻度、严重和不可恢复)合并, 如第3.2节所述。这是合理的, 因为这种方法是用于预警系统, 只需要知道是否有疾病或没有疾病。因此, 将呼吸系统疾病患病率作为一个整体来看, 而不是按严重程度分段, 可以创建一个更简单、更专注于所解决问题挑战的数据集。需要注意的是, 这里开发的方法可以应用于猪的所有类型呼吸系统疾病。

环境条件变化后呼吸系统疾病的检测符合文献中推导的参数, 即我们在病原体潜伏期之前(例如, PRRS的平均潜伏期为14天 [32])不会预测即将到来的疾病。GRU-R和GRU-AE均能够在批次中的猪首先出现症状前1-7天发出即将发生呼吸系统疾病的警报。然而, Luminol模型无法反映这一点, 这可能是由于其无法找到导致强MCC的超参数。这导致PSO最大化 α 和 β 的值, 因为如果性能差, 这样做没有后果。

由于RNN-based模型在自然语言处理等领域表现出色, 其用于序列分析的新兴流行导致需要方法来提高这些黑盒模型的可解释性。该领域的工作已稳步增长 [50,51], 未来的研究可以考虑将此类方法应用于模型, 以基于GRU-AE更好地理解生长猪呼吸系统疾病的风险因素。由于该方法无假设且不考虑国家差异, 额外的工作来验证这种多传感器异常检测是否超越农业健康领域, 从而验证其鲁棒性, 将对该领域社区具有重要价值。

7. 结论

我们提出了一种基于深度学习的模型, 能够对生长猪呼吸系统疾病的早期发生发出预警。所提出的基于GRU的自动编码器与PSO优化异常检测器相结合的方法, 在检测将导致生长猪发生呼吸系统疾病的异常方面显示出强大的潜力, 并且对特定国家环境具有鲁棒性。这种GRU-AE能够优于其他可比方法

in precision, recall, and overall performance in both scenarios where there was a presence or an absence of disease.

Author Contributions: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, and Writing—Original Draft Preparation: J.C.; Data Curation, Funding acquisition, Investigation, Project administration, Resources, Supervision, Validation, Writing Review and Editing: I.K.; Conceptualization, Methodology, Writing—Review and Editing, and Supervision: T.P.; Conceptualization, Methodology, Validation, Writing—Review and Editing, and Supervision: J.B.

Funding: This work was conducted under the PROHEALTH project. PROHEALTH received funding from the European Union Seventh Framework Programme for Research, Technological Development and Demonstration under grant agreement no. 613574.

Acknowledgments: This research made use of the Rocket High Performance Computing service at Newcastle University.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Huhn, R. Swine enzootic pneumonia: Incidence and effect on rate of body weight gain. *Am. J. Vet. Res.* **1970**, *31*, 1097–1108. [PubMed]
2. Stärk, K.D. Epidemiological investigation of the influence of environmental risk factors on respiratory diseases in swine—A literature review. *Vet. J.* **2000**, *159*, 37–56. [CrossRef] [PubMed]
3. Malhotra, P.; Ramakrishnan, A.; Anand, G.; Vig, L.; Agarwal, P.; Shroff, G. LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection. Available online: <https://arxiv.org/abs/1607.00148> (accessed on 20 June 2018).
4. Malhotra, P.; Vig, L.; Shroff, G.; Agarwal, P. Long short term memory networks for anomaly detection in time series. In *Proceedings*; Presses Universitaires de Louvain: Louvain-la-Neuve, Belgium, 2015; p. 89.
5. Morales, I.R.; Cebrián, D.R.; Blanco, E.F.; Sierra, A.P. Early warning in egg production curves from commercial hens: A SVM approach. *Comput. Electron. Agric.* **2016**, *121*, 169–179. [CrossRef]
6. Singh, R.K.; Sharma, V.C. *Ensemble Approach for Zoonotic Disease Prediction Using Machine Learning Techniques*; Indian Institute of Management: Ahmedabad, India, 2015.
7. Valdes-Donoso, P.; VanderWaal, K.; Jarvis, L.S.; Wayne, S.R.; Perez, A.M. Using Machine learning to Predict swine Movements within a regional Program to improve control of infectious Diseases in the US. *Front. Vet. Sci.* **2017**, *4*, 2. [CrossRef] [PubMed]
8. Cabral, G.G.; de Oliveira, A.L.I. One-class Classification for heart disease diagnosis. In Proceedings of the IEEE International Conference on Systems, Man and Cybernetics (SMC), San Diego, CA, USA, 5–8 October 2014; pp. 2551–2556.
9. Leng, Q.; Qi, H.; Miao, J.; Zhu, W.; Su, G. One-class classification with extreme learning machine. *Math. Probl. Eng.* **2015**, 2015. [CrossRef]
10. Thompson, R.; Matheson, S.M.; Plötz, T.; Edwards, S.A.; Kyriazakis, I. Porcine lie detectors: Automatic quantification of posture state and transitions in sows using inertial sensors. *Comput. Electron. Agric.* **2016**, *127*, 521–530. [CrossRef] [PubMed]
11. Münz, G.; Li, S.; Carle, G. Traffic anomaly detection using k-means clustering. In *GI/ITG Workshop MMBnet*, University of Hamburg: Hamburg, Germany, 2017; pp. 13–14.
12. Ambusaidi, M.A.; Tan, Z.; He, X.; Nanda, P.; Lu, L.F.; Jamdagni, A. Intrusion detection method based on nonlinear correlation measure. *Int. J. Internet Protoc. Technol.* **2014**, *8*, 77–86. [CrossRef]
13. Aghabozorgi, S.; Shirkhorshidi, A.S.; Wah, T.Y. Time-series clustering—A decade review. *Inf. Syst.* **2015**, *53*, 16–38. [CrossRef]
14. Wong, W.K.; Moore, A.W.; Cooper, G.F.; Wagner, M.M. Bayesian network anomaly pattern detection for disease outbreaks. In Proceedings of the 20th International Conference on Machine Learning (ICML-03), Washington, DC, USA, 21–24 August 2003; pp. 808–815.
15. Wong, W.K.; Moore, A.; Cooper, G.; Wagner, M. What’s strange about recent events (WSARE): An algorithm for the early detection of disease outbreaks. *J. Mach. Learn. Res.* **2005**, *6*, 1961–1998.

在精确率、召回率和整体性能方面，无论是在存在疾病或不存在疾病的情况下。

作者贡献: 概念化、数据管理、形式分析、调查、方法、软件、验证、可视化以及撰写—原始草稿准备: J.C.; 数据管理、资助获取、调查、项目管理、资源、监督、验证、撰写审阅和编辑: I.K.; 概念化、方法、撰写—审阅和编辑以及监督: T.P.; 概念化、方法、验证、撰写—审阅和编辑以及监督: J.B.

资助: 这项工作是在PROHEALTH项目下进行的。PROHEALTH项目获得了欧盟第七框架计划（用于研究、技术开发和示范）的资助，资助协议号为613574。

致谢: 这项研究使用了纽卡斯尔大学的Rocket高性能计算服务。

利益冲突: 作者声明无利益冲突。

参考文献

1. Huhn, R. 猪地方流行性肺炎：发病率及其对体重增率的影响。美国兽医研究杂志 1970, 31, 1097–1108。 [PubMed]2. Stärk, K.D. 环境风险因素对猪呼吸系统疾病影响的流行病学调查——文献综述。兽医杂志 2000, 159, 37–56。 [CrossRef] [PubMed]3. Malhotra, P.; Ramakrishnan, A.; Anand, G.; Vig, L.; Agarwal, P.; Shroff, G. 基于LSTM的编码器-解码器用于多传感器异常检测。在线获取: <https://arxiv.org/abs/1607.00148>（访问于2018年6月20日）。 4. Malhotra, P.; Vig, L.; Shroff, G.; Agarwal, P. 长短期记忆网络用于时间序列异常检测。载于会议论文集; 鲁汶大学出版社: 比利时鲁汶, 2015; 第89页。 5. Morales, I.R.; Cebrián, D.R.; Blanco, E.F.; Sierra, A.P. 商业蛋鸡产蛋曲线的早期预警：一种支持向量机方法。农业电子计算 2016, 121, 169–179。 [CrossRef]6. Singh, R.K.; Sharma, V.C. 使用机器学习技术进行人畜共患病预测的集成方法; 印度管理学院: 印度艾哈迈达巴德, 2015。 7. Valdes-Donoso, P.; VanderWaal, K.; Jarvis, L.S.; Wayne, S.R.; Perez, A.M. 使用机器学习预测美国区域控制传染病计划内的猪活动。兽医学前沿 2017, 4, 2。 [CrossRef] [PubMed]8. Cabral, G.G.; de Oliveira, A.L.I. 用于心脏病诊断的单类分类。载于IEEE国际系统、人与 cybernetics 会议 (SMC) 论文集, 美国加利福尼亚州圣地亚哥, 2014年10月5日至8日; 第2551–2556页。 9. Leng, Q.; Qi, H.; Miao, J.; Zhu, W.; Su, G. 极限学习机的单类分类。工程数学问题 2015, 2015。 [CrossRef]10. Thompson, R.; Matheson, S.M.; Plötz, T.; Edwards, S.A.; Kyriazakis, I. 猪测谎仪：使用惯性传感器自动量化母猪姿势状态和转换。农业电子计算 2016, 127, 521–530。 [CrossRef] [PubMed]11. Münz, G.; Li, S.; Carle, G. 使用k均值聚类进行交通异常检测。载于GI/ITG Workshop MMBnet, 汉堡大学: 德国汉堡, 2017; 第13–14页。 12. Ambusaidi, M.A.; Tan, Z.; He, X.; Nanda, P.; Lu, L.F.; Jamdagni, A. 基于非线性相关度量的入侵检测方法。国际互联网协议技术杂志 2014, 8, 77–86。 [CrossRef]13. Aghabozorgi, S.; Shirkhorshidi, A.S.; Wah, T.Y. 时间序列聚类——十年综述。信息系统 2015, 53, 16–38。 [CrossRef]14. Wong, W.K.; Moore, A.W.; Cooper, G.F.; Wagner, M.M. 贝叶斯网络异常模式检测用于疾病爆发。载于第20届国际机器学习会议 (ICML-03) 论文集, 美国华盛顿特区, 2003年8月21日至24日; 第808–815页。 15. Wong, W.K.; Moore, A.; Cooper, G.; Wagner, M. 最近事件中的异常 (WSARE)：一种早期检测疾病爆发的算法。机器学习研究杂志 2005, 6, 1961–1998。

16. Chauhan, S.; Vig, L. Anomaly detection in ECG time signals via deep long short-term memory networks. In Proceedings of the IEEE International Conference on Data Science and Advanced Analytics (DSAA), Paris, France, 19–21 October 2015; pp. 1–7.

17. Haque, S.A.; Rahman, M.; Aziz, S.M. Sensor anomaly detection in wireless sensor networks for healthcare. *Sensors* **2015**, *15*, 8764–8786. [CrossRef] [PubMed]

18. Veeravalli, B.; Deepu, C.J.; Ngo, D. Real-Time, Personalized Anomaly Detection in Streaming Data for Wearable Healthcare Devices. In *Handbook of Large-Scale Distributed Computing in Smart Healthcare*; Springer: Berlin, Germany, 2017; pp. 403–426.

19. Kanarachos, S.; Mathew, J.; Chroneos, A.; Fitzpatrick, M. Anomaly detection in time series data using a combination of wavelets, neural networks and Hilbert transform. In Proceedings of the 6th IEEE International Conference on Information, Intelligence, Systems and Applications (IISA2015), Corfu, Greece, 6–8 July 2015; pp. 1–6.

20. Li, J.; Pedrycz, W.; Jamal, I. Multivariate time series anomaly detection: A framework of Hidden Markov Models. *Appl. Soft Comput.* **2017**, *60*, 229–240. [CrossRef]

21. Chuah, M.C.; Fu, F. ECG anomaly detection via time series analysis. In Proceedings of the International Symposium on Parallel and Distributed Processing and Applications, Niagara Falls, ON, Canada, 28 August–1 September 2007; pp. 123–135.

22. Chen, X.W.; Lin, X. Big data deep learning: Challenges and perspectives. *IEEE Access* **2014**, *2*, 514–525. [CrossRef]

23. Kamlaris, A.; Prenafeta-Boldú, F.X. Deep learning in agriculture: A survey. *Comput. Electron. Agric.* **2018**, *147*, 70–90. [CrossRef]

24. Veen, F.V. *A Mostly complete chart of Neural Networks*; The Asimov Institute: Utrecht, The Netherlands, 2016.

25. Lv, Y.; Duan, Y.; Kang, W.; Li, Z.; Wang, F.Y. Traffic flow prediction with big data: A deep learning approach. *IEEE Trans. Intell. Transp. Syst.* **2015**, *16*, 865–873. [CrossRef]

26. Andrews, J.T.; Morton, E.J.; Griffin, L.D. Detecting anomalous data using auto-encoders. *Int. J. Mach. Learn. Comput.* **2016**, *6*, 21.

27. Rahman, A.; Smith, D.; Hills, J.; Bishop-Hurley, G.; Henry, D.; Rawnsley, R. A comparison of autoencoder and statistical features for cattle behaviour classification. In Proceedings of the International Joint Conference on Neural Networks (IJCNN), Vancouver, BC, Canada, 24–29 July 2016; pp. 2954–2960.

28. Al Moubayed, N.; Breckon, T.; Matthews, P.; McGough, A.S. Sms spam filtering using probabilistic topic modelling and stacked denoising autoencoder. In Proceedings of the International Conference on Artificial Neural Networks, Barcelona, Spain, 6–9 September 2016; pp. 423–430.

29. Zoetis. *Individual Pig Care*; Zoetis: Parsippany-Troy Hills, NJ, USA, 2012.

30. Alert, G. Application Areas. Available online: <http://www.general-alert.com/Applications> (accessed on 30 April 2018).

31. Whittemore, C.T.; Kyriazakis, I. *Whittemore’s Science and Practice of Pig Production*; John Wiley & Sons: Hoboken, NJ, USA, 2006.

32. The World Organisation for Animal Health. *Infection With Porcine Reproductive and Respiratory Syndrome Virus*; The World Organisation for Animal Health: Paris, France, 2014.

33. Ioffe, S.; Szegedy, C. Batch normalization: Accelerating deep network training by reducing internal covariate shift. *arXiv* **2015**, preprint arXiv:1502.03167.

34. Sutskever, I.; Vinyals, O.; Le, Q.V. Sequence to sequence learning with neural networks. In *Advances in Neural Information Processing Systems*, Curran Associates, Inc.: New York, NY, USA; 2014; pp. 3104–3112.

35. Tieleman, T.; Hinton, G. Lecture 6.5-rmsprop: Divide the gradient by a running average of its recent magnitude. *COURSERA Neural Netw. Mach. Learn.* **2012**, *4*, 26–31.

36. Cleveland, W.S. Robust locally weighted regression and smoothing scatterplots. *J. Am. Stat. Assoc.* **1979**, *74*, 829–836. [CrossRef]

37. Shi, Y.; Eberhart, R. A modified particle swarm optimizer. In Proceedings of the IEEE International Conference on Evolutionary Computation Proceedings, IEEE World Congress on Computational Intelligence, Anchorage, AK, USA, 4–9 May 1998; pp. 69–73.

38. Holm, S. A simple sequentially rejective multiple test procedure. *Scand. J. Stat.* **1979**, *6*, 65–70.

39. LinkedIn. Luminol. 2016. Available online: <https://github.com/linkedin/luminol> (accessed on 30 April 2018).

16. Chauhan, S.; Vig, L. 异常检测在ECG时间信号中通过深度长短期记忆网络。在IEEE国际会议数据科学与高级分析（DSAA）论文集中，巴黎，法国，2015年10月19日至21日；第1-7页。

17. Haque, S.A.; Rahman, M.; Aziz, S.M. 无线传感器网络中的传感器异常检测用于医疗保健。Sensors 2015, 15, 8764–8786. [CrossRef] [PubMed]

18. Veeravalli, B.; Deepu, C.J.; Ngo, D. 流式数据中可穿戴医疗设备的实时个性化异常检测。在智能医疗保健中大规模分布式计算手册；Springer：柏林，德国，2017年；第403-426页。

19. Kanarachos, S.; Mathew, J.; Chroneos, A.; Fitzpatrick, M. 使用小波、神经网络和Hilbert变换的时间序列数据异常检测。在第六届IEEE信息、智能、系统和应用国际会议（IISA2015）论文集中，科孚，希腊，2015年7月6日至8日；第1-6页。

20. Li, J.; Pedrycz, W.; Jamal, I. 多变量时间序列异常检测：一个隐藏马尔可夫模型框架。Appl. Soft Comput. 2017, 60, 229–240. [CrossRef]

21. Chuah, M.C.; Fu, F. ECG异常检测通过时间序列分析。在国际并行和分布式处理与应用会议论文集中，尼亚加拉瀑布，安大略省，加拿大，2007年8月28日至9月1日；第123-135页。

22. Chen, X.W.; Lin, X. 大数据深度学习：挑战与前景。IEEE Access 2014, 2, 514–525.[CrossRef]

23. Kamlaris, A.; Prenafeta-Boldú, F.X. 农业中的深度学习：一项调查。Comput. Electron. Agric. 2018, 147, 70–90. [CrossRef]

24. Veen, F.V. 神经网络的一个大致完整的图表；The Asimov Institute：乌得勒支，荷兰，2016年。

25. Lv, Y.; Duan, Y.; Kang, W.; Li, Z.; Wang, F.Y. 基于大数据的交通流量预测：一种深度学习方法。IEEE Trans. Intell. Transp. Syst. 2015, 16, 865–873. [CrossRef]

26. Andrews, J.T.; Morton, E.J.; Griffin, L.D. 使用自编码器检测异常数据。Int. J. Mach. Learn. Comput. 2016, 6, 21.

27. Rahman, A.; Smith, D.; Hills, J.; Bishop-Hurley, G.; Henry, D.; Rawnsley, R. 自编码器和统计特征用于牛行为分类的比较。在国际联合神经网络会议（IJCNN）论文集中，温哥华，不列颠哥伦比亚省，加拿大，2016年7月24日至29日；第2954–2960页。

28. Al Moubayed, N.; Breckon, T.; Matthews, P.; McGough, A.S. 使用概率主题建模和堆叠降噪自编码器进行短信垃圾邮件过滤。在国际神经网络会议论文集中，巴塞罗那，西班牙，2016年9月6日至9日；第423–430页。

29. Zoetis. 个体猪护理；Zoetis：帕里斯帕尼 - 特洛伊山，新泽西州，美国，2012 年。

30. Alert, G. 应用领域。在线可用：<http://www.general-alert.com/Applications>（于2018年4月30日访问）。

31. Whittemore, C.T.; Kyriazakis, I. Whittemore的猪生产科学与实践；John Wiley & Sons：霍博肯，新泽西州，美国，2006年。

32. 世界动物卫生组织。感染猪繁殖与呼吸综合征病毒；世界动物卫生组织：巴黎，法国，2014年。

33. Ioffe, S.; Szegedy, C. 批标准化：通过减少内部协变量偏移来加速深度网络训练。arXiv 2015, 预印本 arXiv:1502.03167.

34. Sutskever, I.; Vinyals, O.; Le, Q.V. 使用神经网络的序列到序列学习。在神经信息处理系统进展中；Curran Associates, Inc.：纽约，纽约，美国；2014年；第3104–3112页。

35. Tieleman, T.; Hinton, G. 讲座6.5-rmsprop：将梯度除以其最近幅度的运行平均值。COURSERA神经网络机器学习 2012, 4, 26–31.

36. Cleveland, W.S. 稳健局部加权回归和散点图平滑。J. Am. Stat. Assoc. 1979, 74, 829–836. [CrossRef]

37. Shi, Y.; Eberhart, R. 一种改进的粒子群优化器。在IEEE国际进化计算会议论文集论文中，IEEE计算智能世界大会，安克雷奇，阿拉斯加，美国，1998年5月4日至9日；第69–73页。

38. Holm, S. 一种简单的顺序拒绝多重检验程序。Scand. J. Stat. 1979, 6, 65–70.

39. LinkedIn. Luminol. 2016. 在线可用：<https://github.com/linkedin/luminol>（于2018年4月30日访问）。

40. Lin, J.; Keogh, E.; Wei, L.; Lonardi, S. Experiencing SAX: A novel symbolic representation of time series. *Data Min. Knowl. Discov.* **2007**, *15*, 107–144. [CrossRef]

41. Wei, L.; Kumar, N.; Lolla, V.N.; Keogh, E.J.; Lonardi, S.; Ratanamahatana, C.A. Assumption-Free Anomaly Detection in Time Series. In Proceedings of the International Conference on Scientific and Statistical Database Management, La Jolla, CA, USA, 29 June–1 July 2005; Volume 5, pp. 237–242.

42. Mortensen, S.; Stryhn, H.; Søgaard, R.; Boklund, A.; Stärk, K.D.; Christensen, J.; Willeberg, P. Risk factors for infection of sow herds with porcine reproductive and respiratory syndrome (PRRS) virus. *Prev. Vet. Med.* **2002**, *53*, 83–101. [CrossRef]

43. Chung, J.; Gulcehre, C.; Cho, K.; Bengio, Y. Empirical evaluation of gated recurrent neural networks on sequence modeling. Available online: <https://arxiv.org/abs/1412.3555> (accessed on 12 February 2018).

44. Chung, J.; Gulcehre, C.; Cho, K.; Bengio, Y. Gated feedback recurrent neural networks. In Proceedings of the International Conference on Machine Learning, Lille, France, 6–11 July 2015; pp. 2067–2075.

45. Karpathy, A.; Johnson, J.; Li, F.-F. Visualizing and understanding recurrent networks. Available online: <https://arxiv.org/abs/1506.02078> (accessed on 17 March 2018).

46. Jozefowicz, R.; Zaremba, W.; Sutskever, I. An empirical exploration of recurrent network architectures. In Proceedings of the International Conference on Machine Learning, Lille, France, 6–11 July 2015; pp. 2342–2350.

47. Berckmans, D. Precision livestock farming technologies for welfare management in intensive livestock systems. *Sci. Tech. Re. Off. Int. Epizoot.* **2014**, *33*, 189–196. [CrossRef]

48. Matthews, S.G.; Miller, A.L.; Clapp, J.; Plötz, T.; Kyriazakis, I. Early detection of health and welfare compromises through automated detection of behavioural changes in pigs. *Vet. J.* **2016**, *217*, 43–51. [CrossRef] [PubMed]

49. Chung, Y.; Oh, S.; Lee, J.; Park, D.; Chang, H.H.; Kim, S. Automatic detection and recognition of pig wasting diseases using sound data in audio surveillance systems. *Sensors* **2013**, *13*, 12929–12942. [CrossRef] [PubMed]

50. Krakovna, V.; Doshi-Velez, F. Increasing the interpretability of recurrent neural networks using hidden Markov models. Available online: <https://arxiv.org/abs/1606.05320v2> (accessed on 29 December 2017).

51. Samek, W.; Wiegand, T.; Müller, K.R. Explainable Artificial Intelligence: Understanding, Visualizing and Interpreting Deep Learning Models. Available online: <https://arxiv.org/abs/1708.08296> (accessed on 23 March 2018).



© 2018 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<http://creativecommons.org/licenses/by/4.0/>).

40. 林杰; Keogh, E.; 魏力; Lonardi, S. Experiencing SAX: 一种新的时间序列符号表示方法. 数据挖掘与知识发现 2007, 15, 107–144. [CrossRef]41. 魏力; Kumar, N.; Lolla, V.N.; Keogh, E.J.; Lonardi, S.; Ratanamahatana, C.A. 时间序列无假设异常检测. 在第 5 卷, 第 237–242 页的“国际科学和统计数据库管理会议论文集”, 美国加利福尼亚州拉霍亚, 2005 年 6 月 29 日–7 月 1 日. 42. Mortensen, S.; Stryhn, H.; Søgaard, R.; Boklund, A.; Stärk, K.D.; Christensen, J.; Willeberg, P. 猪繁殖与呼吸综合征 (PRRS) 病毒感染母猪群的风险因素. 兽医预防医学 2002, 53, 83–101. [CrossRef]43. Chung, J.; Gulcehre, C.; Cho, K.; Bengio, Y. Gated Recurrent Neural Network 在序列建模中的经验评估. 在线获取: <https://arxiv.org/abs/1412.3555> (访问于 2018 年 2 月 12 日) . 44. Chung, J.; Gulcehre, C.; Cho, K.; Bengio, Y. Gated Feedback Recurrent Neural Network. 在第 5 卷, 第 2067–2075 页的“国际机器学习会议论文集”, 法国里尔, 2015 年 7 月 6–11 日. 45. Karpathy, A.; Johnson, J.; Li, F.-F. 可视化和理解循环网络. 在线获取: <https://arxiv.org/abs/1506.02078> (访问于 2018 年 3 月 17 日) . 46. Jozefowicz, R.; Zaremba, W.; Sutskever, I. 循环网络架构的经验探索. 在第 5 卷, 第 2342–2350 页的“国际机器学习会议论文集”, 法国里尔, 2015 年 7 月 6–11 日. 47. Berckmans, D. 精准畜牧业技术在密集型畜牧业系统中的福利管理. 国际动物流行病学科学技术报告 2014, 33, 189–196. [CrossRef]48. Matthews, S.G.; Miller, A.L.; Clapp, J.; Plötz, T.; Kyriazakis, I. 通过自动检测猪的行为变化来早期发现健康和福利问题. 兽医杂志 2016, 217, 43–51.[CrossRef] [PubMed]49. Chung, Y.; Oh, S.; Lee, J.; Park, D.; Chang, H.H.; Kim, S. 使用音频监控系统的声音数据自动检测和识别猪的消耗性疾病. 传感器 2013, 13, 12929–12942. [CrossRef] [PubMed]50. Krakovna, V.; Doshi-Velez, F. 使用隐藏马尔可夫模型提高循环神经网络的解释性. 在线获取: <https://arxiv.org/abs/1606.05320v2> (访问于 2017 年 12 月 29 日) . 51. Samek, W.; Wiegand, T.; Müller, K.R. 可解释人工智能: 理解、可视化和解释深度学习模型. 在线获取: <https://arxiv.org/abs/1708.08296> (访问于 2018 年 3 月 23 日) .



© 2018 由作者发表。许可证为MDPI, 瑞士巴塞尔。本文为开放获取文章, 在 Creative Commons Attribution (CC BY)许可协议 (<http://creativecommons.org/licenses/by/4.0/>) 的条款和条件下发布。